

Assignment: MATHEMATICS FOR COMPUTING

Module 2 & 3

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Role of Central Statistical Departments

The statistical system of India is organised at both the Central Government and State Government levels to collect, analyse, interpret and disseminate reliable statistical information for planning and administration. At the central level, statistical activities are mainly coordinated by the Ministry of Statistics and Programme Implementation (MoSPI) through the National Statistical Office (NSO). The central statistical system is responsible for producing major national indicators and maintaining standards for official statistics. Its functions include conducting large-scale sample surveys, compiling National Accounts and Gross Domestic Product (GDP) estimates, collecting information on employment, consumption, industry, prices and other socioeconomic conditions, and developing statistical classifications and methodologies.

These data provide a quantitative basis for national economic and social planning. The Central Government uses official statistics for preparing budgets, formulating economic and welfare policies, monitoring development programmes, estimating employment and unemployment, measuring inflation and evaluating the performance of different sectors of the economy. The NSO also coordinates statistical activities with State Governments and other government agencies to improve the consistency and comparability of data across the country. Official central statistics are further used by researchers, universities, businesses and international organisations for economic analysis and policy studies. Thus, the central statistical system performs a crucial role in transforming large quantities of observations and administrative records into scientifically collected and standardised information that can support evidence-based decision-making throughout India.

Role of State Statistical Departments

At the state level, Kerala has its own statistical agency, the Department of Economics and Statistics (DES), Government of Kerala, which serves as the principal agency for official statistical activities within the state. The department collects, processes, analyses and publishes statistical information relating to Kerala's economy, population and social conditions. Its major responsibilities include preparing estimates of State Income and Gross State Domestic Product (GSDP), collecting and compiling statistics on agriculture, industry, labour, prices, population, trade and other economic activities, and conducting various sample surveys and statistical investigations. The department also provides statistical information required by different Kerala Government departments for planning, budgeting, resource allocation and evaluation of development programmes.

District-level and local-level statistics are particularly important because economic and social conditions vary considerably across different regions of Kerala. Data on agriculture, employment, education, infrastructure, prices and other socioeconomic indicators help administrators identify regional needs and formulate appropriate development strategies. The Kerala department also works in coordination with the central statistical system so that state-level statistics follow appropriate concepts, definitions and methodologies and can be compared with statistics from other states. Therefore, the Central Statistical System provides a national statistical framework, while the Department of Economics and Statistics of Kerala provides detailed state-level statistical information. Together, they form an important foundation for evidence-based governance, economic planning, academic research and data-driven decision-making in India and Kerala.

Coordinate System

Definition

A **coordinate system** provides a numerical framework for representing the position of an object with respect to a chosen reference point and basis.

In a two-dimensional Cartesian coordinate system, a point is represented as

$$P = (x, y),$$

where x and y denote its coordinates along the coordinate axes.

In $\mathbb{R}^n$, a point is represented by

$$\mathbf{x} = (x_1, x_2, \dots, x_n)^T.$$

Practical interpretation

Coordinates are used to represent:

- Position of an object in physics and engineering
- Location of pixels in an image
- Features of a data sample in machine learning

Vectors: Magnitude and Direction

A vector can be represented as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

For a vector in $\mathbb{R}^2$,

$$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}.$$

Its Euclidean magnitude is

$$\|\mathbf{v}\|_2 = \sqrt{v_x^2 + v_y^2}$$

More generally,

$$\|\mathbf{v}\|_2 = \sqrt{\sum_{i=1}^n v_i^2}$$

Displacement as a Vector

Suppose two points are

$$P = (x_1, y_1), \quad Q = (x_2, y_2).$$

The displacement from P to Q is

$$\overrightarrow{PQ} = Q - P = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}$$

Its magnitude is

$$\|\overrightarrow{PQ}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Example

Let

$$P = (1, 2), \quad Q = (5, 5).$$

Then

$$\overrightarrow{PQ} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Vector Operations

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

vector addition is defined componentwise:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

Scalar multiplication is

$$\alpha \mathbf{v} = \begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \\ \alpha v_n \end{bmatrix}$$

From Geometric Vectors to Abstract Vectors

The modern concept of a vector is considerably more general than a physical arrow.

Vector type	Example
Geometric vector	$(3, 4)$
Data vector	(age, income, score)
Image vector	Pixel intensity values
Signal vector	Discrete signal samples
Polynomial	$p(x) = a_0 + a_1x + a_2x^2$
Function	$f(t) = \sin t$
Matrix	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Central idea

What makes these objects “vectors” is not their physical appearance, but whether they satisfy the algebraic axioms of a vector space.

Vector Space: Formal Definition

Let V be a nonempty set and let $\mathbb{F}$ be a field (commonly $\mathbb{R}$ or $\mathbb{C}$).

A set V is called a **vector space over $\mathbb{F}$** if

- ① vectors can be added:

$$+ : V \times V \rightarrow V$$

- ② vectors can be multiplied by scalars:

$$\cdot : \mathbb{F} \times V \rightarrow V$$

- ③ these operations satisfy the vector-space axioms.

Thus,

$$(V, +, \cdot)$$

forms an algebraic structure governed by a fixed collection of axioms.

Vector Space Axioms I

For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\alpha, \beta \in \mathbb{F}$:

1. Closure under addition

$$\mathbf{u} + \mathbf{v} \in V.$$

2. Commutativity of addition

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

3. Associativity of addition

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

4. Additive identity

There exists $\mathbf{0} \in V$ such that

$$\mathbf{u} + \mathbf{0} = \mathbf{u}.$$

Vector Space Axioms II

5. Additive inverse

For every $\mathbf{u} \in V$, there exists $-\mathbf{u} \in V$ such that

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}.$$

6. Closure under scalar multiplication

For $\alpha \in \mathbb{F}$ and $\mathbf{u} \in V$,

$$\alpha \mathbf{u} \in V.$$

7. Distributivity over vector addition

$$\alpha(\mathbf{u} + \mathbf{v}) = \alpha \mathbf{u} + \alpha \mathbf{v}.$$

8. Distributivity over scalar addition

$$(\alpha + \beta)\mathbf{u} = \alpha \mathbf{u} + \beta \mathbf{u}.$$

Vector Space Axioms III

The remaining scalar compatibility axioms are

9. Associativity of scalar multiplication

$$\alpha(\beta \mathbf{u}) = (\alpha\beta)\mathbf{u}.$$

10. Multiplicative identity

$$1\mathbf{u} = \mathbf{u}.$$

Key observation

The vector-space axioms are independent of the physical interpretation of the elements of V .

Therefore, the same theory can describe vectors, functions, polynomials, signals, matrices, and data.

Important Properties Derived from the Axioms

Several useful results follow directly from the vector-space axioms.

Zero scalar

$$0\mathbf{v} = \mathbf{0}$$

for every $\mathbf{v} \in V$.

Scalar multiplication by -1

$$(-1)\mathbf{v} = -\mathbf{v}.$$

Zero vector uniqueness

There exists exactly one additive identity $\mathbf{0}$.

Additive inverse uniqueness

For each $\mathbf{v}$, the vector $-\mathbf{v}$ is unique.

Proof: $0\mathbf{v} = \mathbf{0}$

Using distributivity,

$$(0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}.$$

Since

$$0 + 0 = 0,$$

we obtain

$$0\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}.$$

Adding the additive inverse of $0\mathbf{v}$ to both sides,

$$\boxed{0\mathbf{v} = \mathbf{0}.}$$

Significance

This is not an additional axiom. It is a **theorem derived from the vector-space axioms**.

Examples of Vector Spaces

- ① Euclidean space:

$$\mathbb{R}^n$$

- ② Complex vector space:

$$\mathbb{C}^n$$

- ③ Polynomial space:

$$P_n = \{a_0 + a_1x + \cdots + a_nx^n\}$$

- ④ Space of continuous functions:

$$C[a, b]$$

- ⑤ Matrix space:

$$M_{m \times n}(\mathbb{R})$$

- ⑥ Signal space:

$$\ell^2 = \left\{ (x_1, x_2, \dots) : \sum_{i=1}^{\infty} |x_i|^2 < \infty \right\}.$$

Example: Polynomial Vector Space

Consider

$$P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}.$$

Let

$$p(x) = 1 + 2x + x^2$$

and

$$q(x) = 3 - x + 2x^2.$$

Then

$$p(x) + q(x) = 4 + x + 3x^2 \in P_2.$$

For $\alpha = 2$,

$$2p(x) = 2 + 4x + 2x^2 \in P_2.$$

Hence P_2 is closed under vector addition and scalar multiplication. The polynomial

$$p(x) = a + bx + cx^2$$

can therefore be treated as a vector:

$$\begin{bmatrix} a \\ b \end{bmatrix}.$$

Example: Function Space

Consider the set

$$C[0, 1] = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}.$$

For

$$f(x) = x^2, \quad g(x) = \sin x,$$

we have

$$(f + g)(x) = x^2 + \sin x.$$

For any scalar α ,

$$(\alpha f)(x) = \alpha x^2.$$

Both remain continuous on $[0, 1]$.

Therefore,

$C[0, 1] \text{ is a vector space over } \mathbb{R}.$

Application

Function spaces form the mathematical foundation of differential equations, Fourier analysis, signal processing and functional analysis.

Non-Example of a Vector Space

Consider

$$S = \{(x, y) \in \mathbb{R}^2 : x + y = 1\}.$$

Take

$$\mathbf{u} = (1, 0), \quad \mathbf{v} = (0, 1).$$

Both belong to S because

$$1 + 0 = 1, \quad 0 + 1 = 1.$$

However,

$$\mathbf{u} + \mathbf{v} = (1, 1),$$

and

$$1 + 1 = 2 \neq 1.$$

Thus

$$\mathbf{u} + \mathbf{v} \notin S.$$

Therefore,

S is not a vector space.

Geometric Interpretation of Vector Addition

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

their sum is

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}.$$

Geometrically, vector addition can be represented using the **parallelogram rule**.

$$\mathbf{u} + \mathbf{v} \iff \text{diagonal of the parallelogram}$$

This geometric interpretation is useful, but the algebraic axioms allow vector addition to be generalized to abstract spaces.

Practical Example: Data Vectors

Suppose a student is represented by the feature vector

$$\mathbf{x} = \begin{bmatrix} \text{study hours} \\ \text{attendance} \\ \text{assignment score} \\ \text{test score} \end{bmatrix}.$$

For example,

$$\mathbf{x} = \begin{bmatrix} 8 \\ 90 \\ 85 \\ 88 \end{bmatrix}.$$

A dataset containing N students can therefore be represented as

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_N^T \end{bmatrix}.$$

Generalization in Linear Algebra

The progression can be summarized as

Coordinate $\rightarrow$ Vector $\rightarrow$ Vector Space $\rightarrow$ Linear Transformation

- A coordinate system gives numerical representation.
- A vector represents an element with linear operations.
- A vector space provides the complete algebraic structure.
- A linear transformation maps vectors between vector spaces while preserving linear combinations.

For a linear transformation

$$T : V \rightarrow W,$$

we require

$$T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}).$$

This becomes the foundation for matrices, eigenvalues, differential equations, optimization and machine learning.

Summary

Core concepts

- Coordinates provide a numerical representation of position.
- Vectors encode magnitude, direction and generalized data.
- Vector spaces provide an abstract framework for vectors.
- Closure, associativity, commutativity, identity, inverse and distributive properties define the algebraic structure.

Key equation

$$\alpha(\mathbf{u} + \mathbf{v}) + \beta\mathbf{w} = \alpha\mathbf{u} + \alpha\mathbf{v} + \beta\mathbf{w}$$

demonstrates how linear combinations operate inside a vector space.

Main idea

The power of linear algebra lies in the fact that one set of axioms can describe geometric vectors, data, images, functions, polynomials and signals within a single mathematical framework.

Vector Addition

Definition

Vector addition is the operation of combining two or more vectors by adding their corresponding components.

Let

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

Their sum is defined componentwise as

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

Thus, vector addition combines corresponding components of the vectors.

Example of Vector Addition

Consider two vectors in $\mathbb{R}^2$:

$$\mathbf{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

Their sum is

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 3 + 1 \\ 2 + 4 \end{bmatrix} = \boxed{\begin{bmatrix} 4 \\ 6 \end{bmatrix}}$$

Therefore,

$$\boxed{\mathbf{u} + \mathbf{v} = (4, 6)}$$

Interpretation

The resulting vector represents the combined effect of the two original vectors.

Geometric Interpretation

Vector addition has two important geometric interpretations:

- 1 **Head-to-tail method**
- 2 **Parallelogram method**

Head-to-tail method

The initial point of $\mathbf{v}$ is placed at the terminal point of $\mathbf{u}$. The vector from the initial point of $\mathbf{u}$ to the terminal point of $\mathbf{v}$ represents

$$\mathbf{u} + \mathbf{v}.$$

Parallelogram method

When two vectors originate from the same point, their sum is represented by the diagonal of the parallelogram formed by the two vectors.

Properties of Vector Addition

For all

$$\mathbf{u}, \mathbf{v}, \mathbf{w} \in V,$$

vector addition satisfies the following properties.

1. Closure

$$\mathbf{u} + \mathbf{v} \in V$$

The sum of two vectors in V is also an element of V .

2. Commutativity

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

The order of addition does not affect the result.

3. Associativity

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

The grouping of vectors does not affect the result.

Identity and Inverse under Vector Addition

Additive Identity

Every vector space contains a unique zero vector $\mathbf{0}$ satisfying

$$\mathbf{u} + \mathbf{0} = \mathbf{u}.$$

For example,

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Additive Inverse

For every vector $\mathbf{u}$, there exists a vector $-\mathbf{u}$ such that

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}.$$

If

$$\mathbf{u} = \begin{bmatrix} 2 \\ -5 \end{bmatrix},$$

then

$$-\mathbf{u} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}.$$

Vector Addition as a Binary Operation

In an abstract vector space V , addition is a function

$$+ : V \times V \rightarrow V$$

This means that every ordered pair

$$(\mathbf{u}, \mathbf{v}) \in V \times V$$

is associated with exactly one vector

$$\mathbf{u} + \mathbf{v} \in V.$$

The closure property is therefore essential:

$$\mathbf{u}, \mathbf{v} \in V \implies \mathbf{u} + \mathbf{v} \in V.$$

Linear Algebra Perspective

Vector addition is not restricted to geometric arrows. The same operation can be defined for polynomials, functions, matrices, signals and data vectors, provided the vector-space axioms are satisfied.

Generalization Beyond $\mathbb{R}^2$

For vectors in $\mathbb{R}^n$,

$$\mathbf{u} = (u_1, u_2, \dots, u_n), \quad \mathbf{v} = (v_1, v_2, \dots, v_n),$$

vector addition is

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n).$$

For example,

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 10 \\ 12 \end{bmatrix}.$$

Thus, the geometric interpretation in $\mathbb{R}^2$ is only a special case of a general algebraic operation.

Vector Addition of Polynomials

Consider the polynomial vectors

$$p(x) = 1 + 2x + x^2$$

and

$$q(x) = 3 - x + 4x^2.$$

Their sum is

$$\begin{aligned}(p + q)(x) &= (1 + 2x + x^2) + (3 - x + 4x^2) \\ &= 4 + x + 5x^2.\end{aligned}$$

Therefore,

$$p(x) + q(x) = 4 + x + 5x^2.$$

This demonstrates that vector addition does not require the vectors to be numerical coordinate tuples.

Practical Example: Displacement

Suppose an object undergoes two successive displacements:

$$\mathbf{d}_1 = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \text{ m}$$

and

$$\mathbf{d}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \text{ m.}$$

The total displacement is

$$\begin{aligned} \mathbf{d} &= \mathbf{d}_1 + \mathbf{d}_2 \\ &= \begin{bmatrix} 4 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \boxed{\begin{bmatrix} 6 \\ 2 \end{bmatrix} \text{ m.}} \end{aligned}$$

The magnitude of the resultant displacement is

$$\|\mathbf{d}\| = \sqrt{6^2 + 2^2} = \sqrt{40} = 2\sqrt{10} \text{ m.}$$

Practical Example: Forces

Suppose two forces act on an object:

$$\mathbf{F}_1 = \begin{bmatrix} 10 \\ 5 \end{bmatrix} \text{ N}, \quad \mathbf{F}_2 = \begin{bmatrix} -4 \\ 8 \end{bmatrix} \text{ N}.$$

The resultant force is

$$\begin{aligned} \mathbf{F}_R &= \mathbf{F}_1 + \mathbf{F}_2 \\ &= \begin{bmatrix} 10 - 4 \\ 5 + 8 \end{bmatrix} = \boxed{\begin{bmatrix} 6 \\ 13 \end{bmatrix} \text{ N}}. \end{aligned}$$

Its magnitude is

$$\boxed{\|\mathbf{F}_R\| = \sqrt{6^2 + 13^2} = \sqrt{205} \text{ N}.$$

Vector Addition in Data Science

Let two observations be represented by feature vectors

$$\mathbf{x} = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}.$$

Their component-wise combination is

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} 3 \\ 8 \\ 11 \end{bmatrix}.$$

In data analysis, vector operations are used in

- Feature representation
- Signal processing
- Image processing
- Machine learning
- Optimization

The same algebraic operation therefore connects physical systems with computational data

Key Result

Vector addition satisfies

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$\mathbf{u} + \mathbf{0} = \mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$$

and

$$\mathbf{u}, \mathbf{v} \in V \Rightarrow \mathbf{u} + \mathbf{v} \in V.$$

Central Idea

Vector addition is a closed binary operation that provides a unified mathematical mechanism for combining geometric quantities, physical effects, data, functions, polynomials and other abstract vectors.

Summary

- Vector addition combines corresponding components.
- Geometrically, it can be visualized using the head-to-tail or parallelogram method.
- Algebraically, addition is a binary operation

$$+ : V \times V \rightarrow V.$$

- It satisfies closure, commutativity and associativity.
- Every vector has an additive identity and an additive inverse.
- The concept extends naturally from $\mathbb{R}^2$ to abstract vector spaces.
- Applications include displacement, forces, velocity, signals, images and machine-learning data.

Geometric addition $\longrightarrow$ Algebraic addition $\longrightarrow$ Abstract vector space

Scalar Multiplication

Definition

Scalar multiplication is the operation of multiplying every component of a vector by a scalar.

Let

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

and let $\alpha \in \mathbb{R}$ be a scalar.

Then scalar multiplication is defined as

$$\alpha \mathbf{v} = \begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \\ \alpha v_n \end{bmatrix}$$

Example of Scalar Multiplication

Consider

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

For $\alpha = 3$,

$$3\mathbf{v} = 3 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \end{bmatrix}.$$

For $\alpha = -2$,

$$-2\mathbf{v} = \begin{bmatrix} -4 \\ -6 \end{bmatrix}.$$

Therefore,

$$\boxed{\alpha\mathbf{v}}$$

changes the size and, depending on the sign of α , possibly the direction of the vector.

Effect on Magnitude and Direction

For a vector $\mathbf{v}$ and scalar α ,

$$\|\alpha\mathbf{v}\| = |\alpha| \|\mathbf{v}\|$$

Therefore:

Scalar	Effect
$\alpha > 1$	Magnitude increases
$0 < \alpha < 1$	Magnitude decreases
$\alpha = 1$	Vector remains unchanged
$\alpha = 0$	Vector becomes zero vector
$\alpha < 0$	Direction reverses
$\alpha = -1$	Same magnitude, opposite direction

Hence scalar multiplication controls the **scale and orientation** of a vector.

Geometric Interpretation

Let

$$\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Then

$$2\mathbf{v} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

has the same direction but twice the magnitude. Similarly,

$$\frac{1}{2}\mathbf{v} = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}$$

has the same direction but half the magnitude. For a negative scalar,

$$-2\mathbf{v} = \begin{bmatrix} -4 \\ -2 \end{bmatrix},$$

which has twice the magnitude and points in the opposite direction.

Geometric Principle

Scalar multiplication preserves the line through the origin containing $\mathbf{v}$.

Scalar Multiplication in an Abstract Vector Space

In an abstract vector space V over a field $\mathbb{F}$,

$$\cdot : \mathbb{F} \times V \rightarrow V$$

is the scalar multiplication operation.

For

$$\alpha \in \mathbb{F}, \quad \mathbf{v} \in V,$$

the result

$$\alpha \mathbf{v}$$

must also belong to V .

Therefore, the closure property is

$$\alpha \mathbf{v} \in V.$$

The scalar need not represent a physical quantity such as length. It may simply be an element of the underlying field.

Axioms of Scalar Multiplication

For all $\alpha, \beta \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$:

1. Distributivity over vector addition

$$\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$$

2. Distributivity over scalar addition

$$(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$$

3. Compatibility of scalar multiplication

$$\alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v}$$

4. Multiplicative identity

$$1\mathbf{v} = \mathbf{v}$$

Important Derived Properties

The axioms imply several important identities.

Multiplication by zero

$$0\mathbf{v} = \mathbf{0}$$

Multiplication by -1

$$(-1)\mathbf{v} = -\mathbf{v}$$

Multiplication by $-\alpha$

$$(-\alpha)\mathbf{v} = -(\alpha\mathbf{v})$$

Zero vector

$$\alpha\mathbf{0} = \mathbf{0}$$

for every scalar α .

Proof: $\alpha \mathbf{0} = \mathbf{0}$

Using the fact that

$$\mathbf{0} + \mathbf{0} = \mathbf{0},$$

we have

$$\alpha \mathbf{0} = \alpha(\mathbf{0} + \mathbf{0}).$$

By distributivity,

$$\alpha \mathbf{0} = \alpha \mathbf{0} + \alpha \mathbf{0}.$$

Subtracting $\alpha \mathbf{0}$ from both sides gives

$$\boxed{\alpha \mathbf{0} = \mathbf{0}.}$$

Observation

This property is a consequence of the vector-space axioms and is not an independent axiom.

Linear Combination

Scalar multiplication becomes especially important when combined with vector addition.

A vector of the form

$$\mathbf{w} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_k \mathbf{v}_k$$

is called a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_k$.

The scalars

$$\alpha_1, \alpha_2, \dots, \alpha_k$$

are called the **coefficients**.

Importance

Linear combinations are the foundation of:

span $\rightarrow$ linear independence $\rightarrow$ basis $\rightarrow$ dimension.

Example: Linear Combination

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Consider

$$2\mathbf{v}_1 - \mathbf{v}_2.$$

Then

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} -1 \\ 3 \end{bmatrix}}.$$

Therefore,

$$\begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Span and Scalar Multiplication

Given vectors

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k,$$

their span is

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \left\{ \sum_{i=1}^k \alpha_i \mathbf{v}_i : \alpha_i \in \mathbb{F} \right\}.$$

Thus scalar multiplication allows each vector to be scaled, while vector addition combines the scaled vectors.

Scalar multiplication + Vector addition = Linear combination

This is one of the central constructions in linear algebra.

Practical Example: Physical Modeling

Suppose the velocity of an object is

$$\mathbf{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \text{ m/s.}$$

After 2 seconds, assuming constant velocity,

$$\mathbf{d} = 2\mathbf{v} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} \text{ m.}$$

The magnitude is

$$\|\mathbf{d}\| = \sqrt{8^2 + 6^2} = 10\text{m.}$$

Hence scalar multiplication provides a direct mathematical model for scaling a physical quantity with time.

Practical Example: Machine Learning

In machine learning, an input sample can be represented by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

A model may scale features using

$$\mathbf{x}' = \alpha \mathbf{x}.$$

More generally, weighted feature combinations take the form

$$y = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

or

$$y = \mathbf{w}^T \mathbf{x}.$$

Here the weights w_i act as scalars on individual components.

Scalar Multiplication of Polynomials

Let

$$p(x) = 1 + 2x + 3x^2.$$

For scalar $\alpha = 4$,

$$4p(x) = 4(1 + 2x + 3x^2)$$

$$4p(x) = 4 + 8x + 12x^2.$$

Therefore, scalar multiplication is also meaningful in polynomial vector spaces.

For the polynomial vector space P_n ,

$$\alpha p(x) \in P_n$$

for every $\alpha \in \mathbb{R}$ and $p(x) \in P_n$.

Scalar Multiplication and Linear Transformations

Scalar multiplication is preserved by every linear transformation

$T : V \rightarrow W$:

$$T(\alpha \mathbf{v}) = \alpha T(\mathbf{v}).$$

Together with

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}),$$

we obtain

$$T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}).$$

This property explains why scalar multiplication is fundamental to linear transformations and matrix representations.

Summary

- Scalar multiplication multiplies every component of a vector by a scalar.
- The magnitude satisfies

$$\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|.$$

- A negative scalar reverses the vector's direction.
- In an abstract vector space,

$$\cdot : \mathbb{F} \times V \rightarrow V.$$

- Scalar multiplication satisfies distributive and compatibility axioms.
- Together with vector addition, it produces linear combinations.
- Linear combinations lead naturally to span, basis, dimension and linear transformations.

Scalar multiplication $\rightarrow$ Linear combinations $\rightarrow$ Span and Basis
--

Linear Combination

Definition

A **linear combination** of a collection of vectors is obtained by multiplying each vector by a scalar and then adding the resulting vectors.

Let

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in V$$

and let

$$\alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{F}.$$

Then

$$\mathbf{w} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_k \mathbf{v}_k$$

is called a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_k$.

The scalars $\alpha_1, \dots, \alpha_k$ are called **coefficients** or **weights**.

Linear Combination in Matrix Form

Let

$$A = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_k \\ | & | & \cdots & | \end{bmatrix}$$

and define the coefficient vector

$$\mathbf{c} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_k \end{bmatrix}.$$

Then the linear combination

$$\mathbf{w} = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k$$

can be written compactly as

$$\mathbf{w} = A\mathbf{c}.$$

Thus, finding a linear combination is equivalent to multiplying a matrix by a coefficient vector.

Important connection

Linear combinations provide the conceptual bridge between vectors and matrix multiplication.

Example of a Linear Combination

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Consider the linear combination

$$\mathbf{w} = 2\mathbf{u} - \mathbf{v}.$$

Then

$$\begin{aligned} \mathbf{w} &= 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 3 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\begin{bmatrix} -1 \\ 3 \end{bmatrix} = 2\mathbf{u} - \mathbf{v}$$

is a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

Geometric Interpretation

For two vectors

$$\mathbf{u}, \mathbf{v} \in \mathbb{R}^2,$$

a general linear combination has the form

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v}.$$

The scalars a and b independently scale the two vectors.

Therefore,

$$a\mathbf{u}$$

and

$$b\mathbf{v}$$

can be added to produce a new vector $\mathbf{w}$.

If $\mathbf{u}$ and $\mathbf{v}$ are linearly independent, their linear combinations can generate every vector in $\mathbb{R}^2$.

$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \mathbb{R}^2$$

when $\mathbf{u}$ and $\mathbf{v}$ are linearly independent.

Interactive Geometric Interpretation

For

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 2 \end{bmatrix},$$

a general linear combination is

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v}.$$

The coefficients a and b determine the contribution of each vector.

The set of all possible endpoints obtained by varying $a, b \in \mathbb{R}$ forms the span of $\mathbf{u}$ and $\mathbf{v}$.

Properties of Linear Combinations

Suppose

$$\mathbf{w} = \sum_{i=1}^k \alpha_i \mathbf{v}_i.$$

Important properties include:

Closure

If $\mathbf{v}_i \in V$ and $\alpha_i \in \mathbb{F}$, then

$$\mathbf{w} \in V.$$

Zero linear combination

Choosing

$$\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$$

gives

$$\mathbf{w} = \mathbf{0}.$$

Span of a Set of Vectors

The **span** of vectors

$$\mathbf{v}_1, \dots, \mathbf{v}_k$$

is the set of all possible linear combinations:

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \left\{ \sum_{i=1}^k \alpha_i \mathbf{v}_i : \alpha_i \in \mathbb{F} \right\}.$$

Therefore,

$$\text{span}(S) = \{\text{all linear combinations of elements of } S\}.$$

The span is always a subspace of the vector space V .

Key idea

Linear combination describes how vectors are constructed; span describes the complete collection of vectors that can be constructed.

Example: Span in $\mathbb{R}^2$

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

A general linear combination is

$$a\mathbf{v}_1 + b\mathbf{v}_2 = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Since a, b can be any real numbers,

$$\boxed{\text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2.}$$

These two vectors form the standard basis of $\mathbb{R}^2$.

Linearly Dependent Vectors

Consider

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Observe that

$$\mathbf{v}_2 = 2\mathbf{v}_1.$$

Therefore,

$$a\mathbf{v}_1 + b\mathbf{v}_2 = (a + 2b)\mathbf{v}_1.$$

Thus, every linear combination lies on the same line.

$$\boxed{\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{span}\{\mathbf{v}_1\}.$$

Hence two vectors do not necessarily generate a two-dimensional space.

Insight

The dimension generated by a collection of vectors depends on their linear independence, not simply on their number.

Linear Independence and Linear Combination

Vectors

$$\mathbf{v}_1, \dots, \mathbf{v}_k$$

are **linearly independent** if

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0}$$

implies

$$\alpha_1 = \alpha_2 = \dots = \alpha_k = 0.$$

If a nontrivial set of coefficients exists, then the vectors are linearly dependent.

Therefore, linear combinations provide the mathematical mechanism for defining linear independence.

Linear Combination and Basis

A set

$$B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$$

is a **basis** for a vector space V if:

- ① Spanning: The vectors in B can produce every vector in V ..
- ② Linear independence: None of the vectors in B is redundant..

Equivalently, every vector $\mathbf{x} \in V$ can be written uniquely as

$$\mathbf{x} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n.$$

The coefficients α_i are the coordinates of $\mathbf{x}$ relative to the basis B .

Linear combination $\rightarrow$ Span + Linear independence $\rightarrow$ Basis

Practical Example: Linear Regression

Suppose a prediction model uses two features x_1 and x_2 :

$$\hat{y} = w_1x_1 + w_2x_2 + b.$$

Ignoring the intercept for the moment,

$$\hat{y} = w_1x_1 + w_2x_2$$

is a linear combination of the feature values.

In matrix notation,

$$\hat{y} = \mathbf{w}^T \mathbf{x}.$$

For many observations,

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}.$$

Thus linear regression can be interpreted as finding the appropriate linear combination of feature vectors to approximate the target data.

Practical Example: Image Processing

Suppose an image is represented by vectors

$$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k.$$

A new image approximation can be expressed as

$$\mathbf{x} \approx \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \dots + \alpha_k \mathbf{x}_k.$$

The coefficients determine how much each component image contributes to the reconstructed image.

This idea appears in:

- Principal Component Analysis
- Image compression
- Eigenfaces
- Feature extraction
- Low-dimensional representation

Linear Combination in Function Spaces

The concept extends beyond finite-dimensional vectors.

Consider functions

$$f_1(x) = 1, \quad f_2(x) = x, \quad f_3(x) = x^2.$$

A linear combination is

$$f(x) = \alpha_1 + \alpha_2 x + \alpha_3 x^2.$$

Thus the polynomial

$$3 - 2x + 5x^2$$

is a linear combination of

$$1, \quad x, \quad x^2.$$

Therefore,

$$P_2 = \text{span}\{1, x, x^2\}.$$

Linear Combination and Matrix Columns

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}.$$

Its columns are

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix},$$

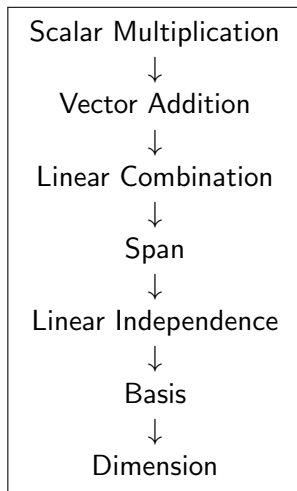
matrix multiplication gives

$$A\mathbf{x} = c_1\mathbf{a}_1 + c_2\mathbf{a}_2.$$

Therefore

Fundamental Linear Algebra Connection

Linear combinations form a central chain of ideas:



These concepts provide the structural foundation of finite-dimensional and infinite-dimensional linear algebra.

Summary

- A linear combination is

$$\mathbf{w} = \sum_{i=1}^k \alpha_i \mathbf{v}_i.$$

- The scalars α_i determine the contribution of each vector.
- The set of all possible linear combinations is the span.
- Linear dependence and independence are defined using linear combinations.
- A basis is a linearly independent spanning set.
- Matrix multiplication is fundamentally a linear combination of matrix columns.
- Linear combinations appear in regression, PCA, image processing, signal processing and machine learning.

Linear combinations are the building blocks of linear algebra.

5.1 Definition of Linear Independence

Definition

Let

$$\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

be a set of vectors in a vector space V .

The vectors are said to be **linearly independent** if

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0}$$

has only the **trivial solution**

$$\alpha_1 = \alpha_2 = \dots = \alpha_k = 0$$

Otherwise, the vectors are called **linearly dependent**.

Interpretation:

- Independent vectors contain no redundant direction.
- Dependent vectors contain at least one vector that can be expressed as a linear combination of the others.

5.2 Linear Dependence

A set of vectors is linearly dependent if there exist scalars $\alpha_1, \dots, \alpha_k$, **not all zero**, such that

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0}$$

For example,

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Since

$$\mathbf{v}_2 = 2\mathbf{v}_1,$$

we obtain

$$2\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}.$$

The coefficients $(2, -1)$ are not both zero.

Therefore,

$\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent.

5.3 Linear Independence: Numerical Example

Consider

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

To test independence, solve

$$a\mathbf{u} + b\mathbf{v} = \mathbf{0}.$$

Thus,

$$a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This gives

$$\begin{cases} a + 3b = 0, \\ 2a + b = 0. \end{cases}$$

From the first equation,

$$a = -3b.$$

Substituting into the second,

$$2(-3b) + b = -5b = 0.$$

Hence

$$b = 0, \quad a = 0.$$

Therefore,

$\mathbf{u}, \mathbf{v}$ are linearly independent.
--

5.4 Matrix Test for Linear Independence

Place the vectors as columns of a matrix:

$$A = \left[\begin{array}{c|c|c|c} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_k \\ | & | & & | \end{array} \right] .$$

The linear independence condition becomes

$$A\mathbf{c} = \mathbf{0},$$

where

$$\mathbf{c} = \left[\begin{array}{c} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_k \end{array} \right] .$$

Therefore,

$$\text{Vectors are independent} \iff \text{Null}(A) = \{\mathbf{0}\}$$

Equivalently,

$$\text{rank}(A) = k$$

when A has k columns.

For a square matrix,

$$\det(A) \neq 0 \iff \text{columns of } A \text{ are linearly independent.}$$

5.5 Geometric Interpretation

For two vectors in $\mathbb{R}^2$:

Dependent vectors

$$\mathbf{v}_2 = c\mathbf{v}_1$$

lie on the same line through the origin.

They therefore provide only

one fundamental direction

and cannot generate the entire plane.

Independent vectors

point in different directions.

Their linear combinations

$$a\mathbf{v}_1 + b\mathbf{v}_2$$

can generate every vector in $\mathbb{R}^2$.

Thus,

5.6 Determinant and Geometric Independence

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

form

$$A = \begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix}.$$

Then

$$\det(A) = u_1 v_2 - u_2 v_1.$$

The absolute determinant represents the area of the parallelogram generated by the vectors:

$\text{Area} = |\det(A)|$

Hence,

5.7 Dependence vs Independence

Property	Dependent	Independent
Zero combination	Non-trivial solution exists	Only trivial solution
Redundancy	Yes	No
Directions	At least one redundant	Fundamentally distinct
Determinant (2×2)	0	$\neq 0$
Parallelogram area	0	> 0
Information	Redundant	Non-redundant

The key question is:

Does any vector provide genuinely new information?

5.8 Connection with Information

Linear independence has an important interpretation in data science. Suppose two features are

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

Since

$$x_2 = 2x_1,$$

we have

$$x_2 - 2x_1 = 0.$$

Therefore the features are linearly dependent.

The second feature does not introduce a new direction in feature space. Hence it is **redundant from a linear-algebra perspective**.

Dependence $\longrightarrow$ redundant information

5.9 Multicollinearity in Regression

Consider a linear regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon.$$

If

$$x_2 \approx cx_1,$$

then the predictor columns of the design matrix are nearly linearly dependent.

The matrix

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \end{bmatrix}$$

becomes nearly rank-deficient.

This is known as

multicollinearity.

5.10 Connection to Basis

The concepts introduced so far form a natural progression:

Linear Combination $\rightarrow$ Span $\rightarrow$ Linear Independence $\rightarrow$ Basis $\rightarrow$ Dimension

A set

$$B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$$

is a **basis** of a vector space V if:

- 1 B spans V ;
- 2 B is linearly independent.

Thus,

Basis = Spanning set + Linear independence

Independence ensures that no vector in the basis is redundant.

5.11 Standard Basis of $\mathbb{R}^2$

Consider

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Every vector

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

can be written as

$$\mathbf{x} = x\mathbf{e}_1 + y\mathbf{e}_2.$$

Therefore,

$$\text{span}\{\mathbf{e}_1, \mathbf{e}_2\} = \mathbb{R}^2.$$

To test independence,

$$a\mathbf{e}_1 + b\mathbf{e}_2 = \mathbf{0}$$

5.12 Dimension

The number of vectors in a basis is called the **dimension** of the vector space.

For $\mathbb{R}^2$,

$$\dim(\mathbb{R}^2) = 2.$$

For $\mathbb{R}^3$,

$$\dim(\mathbb{R}^3) = 3.$$

More generally,

$$\dim(V) = \text{number of vectors in any basis of } V$$

A fundamental theorem of linear algebra states that every basis of a finite-dimensional vector space contains the same number of vectors.

Therefore, independence is directly connected to the structure and dimensionality of a vector space.

5.13 Example in $\mathbb{R}^3$

Consider

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Notice that

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2.$$

Therefore,

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}.$$

The coefficients

$$(1, 1, -1)$$

are not all zero.

Hence,

5.14 Function-Space Generalization

Linear independence is not restricted to geometric vectors.
Consider the functions

$$f_1(x) = 1, \quad f_2(x) = x, \quad f_3(x) = x^2.$$

Suppose

$$a + bx + cx^2 = 0$$

for every x .

A polynomial is identically zero only when all its coefficients are zero.
Therefore,

$$a = b = c = 0.$$

Hence,

$\{1, x, x^2\}$ is linearly independent.

In fact,

5.15 Linear Independence in General Vector Spaces

Let V be any vector space over a field $\mathbb{F}$.

The set

$$\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq V$$

is linearly independent if

$$\sum_{i=1}^k \alpha_i \mathbf{v}_i = \mathbf{0}$$

implies

$$\alpha_i = 0 \quad \text{for all } i.$$

This definition applies to:

- $\mathbb{R}^n$,
- polynomial spaces,
- matrix spaces,
- function spaces,
- solution spaces of differential equations

5.16 Application: Dimensionality Reduction

High-dimensional datasets may contain strongly correlated or dependent features.

For example,

$$x_3 \approx 2x_1 + x_2.$$

Then x_3 contributes little new information.

Dimensionality reduction methods attempt to represent the data using fewer effective directions.

One important example is **Principal Component Analysis (PCA)**.

PCA constructs orthogonal directions

$$\mathbf{w}_1, \mathbf{w}_2, \dots$$

that capture decreasing amounts of variance.

The goal is to obtain a lower-dimensional representation

$$X \approx ZW^T$$

while retaining as much relevant information as possible.

5.17 Worked Summary Example

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Step 1: Check dependence

$$\mathbf{v}_2 = 2\mathbf{v}_1.$$

Therefore,

$$\mathbf{v}_2 - 2\mathbf{v}_1 = \mathbf{0}.$$

Step 2: Identify redundancy

One vector can be constructed from the other.

Hence the second vector adds no new direction.

Step 3: Determine the span

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{span}\{\mathbf{v}_1\}.$$

Thus the span is only a line through the origin.

5.18 Key Mathematical Results

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0}$$

has only the trivial solution

$$\alpha_1 = \cdots = \alpha_k = 0$$

$\implies$ **linear independence.**

For a matrix A with columns $\mathbf{v}_1, \dots, \mathbf{v}_k$:

$$\text{Null}(A) = \{\mathbf{0}\}$$

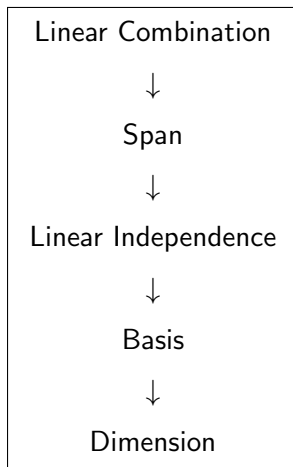
$$\text{rank}(A) = k$$

and for a square matrix,

$$\det(A) \neq 0.$$

These are equivalent ways of expressing independence.

5.19 Conceptual Connection



Linear combination: How can vectors be constructed?

Span: What set of vectors can be constructed?

Independence: Are any directions redundant?

5.20 Applications of Linear Independence

Linear independence is fundamental in:

- **Linear regression:** detecting multicollinearity.
- **Machine learning:** identifying redundant features.
- **Feature selection:** removing unnecessary variables.
- **PCA:** finding effective lower-dimensional directions.
- **Numerical analysis:** understanding rank and conditioning.
- **Signal processing:** separating independent components.
- **Differential equations:** identifying independent solutions.
- **Linear transformations:** determining rank and nullity.

Independence $\implies$ No redundancy $\implies$ Efficient representation

5.21 Final Summary

Linear Independence describes whether a collection of vectors contains redundant directions.

- Only the trivial solution $\Rightarrow$ independent.
- Non-trivial zero combination $\Rightarrow$ dependent.
- Dependent vectors contain redundant information.
- Independent vectors provide fundamentally distinct directions.
- Independence can be tested using rank, null space or determinant.
- A basis is a spanning and linearly independent set.
- The number of basis vectors determines dimension.

Linear Independence $\rightarrow$ Basis $\rightarrow$ Dimension

6. Matrix Multiplication

Definition

Let

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p}.$$

The matrix product AB is an $m \times p$ matrix whose (i, j) -th entry is

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

- The number of columns of A must equal the number of rows of B .
- The resulting matrix has the dimensions

$$(m \times n)(n \times p) = m \times p.$$

- Each entry is obtained from the dot product of a row of A with a column of B .

Row-Column Interpretation

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}.$$

Central computational rule

$$(AB)_{ij} = \text{Row}_i(A) \cdot \text{Column}_j(B)$$

For example,

$$(AB)_{12} = a_{11}b_{12} + a_{12}b_{22} + \cdots + a_{1n}b_{n2}.$$

Worked Example: Matrix Multiplication

Consider

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1(2) + 2(0) & 1(1) + 2(3) \\ 3(2) + 4(0) & 3(1) + 4(3) \end{bmatrix}.$$

Therefore,

$$AB = \begin{bmatrix} 2 & 7 \\ 6 & 15 \end{bmatrix}$$

Similarly,

$$BA = \begin{bmatrix} 5 & 8 \\ 9 & 12 \end{bmatrix}.$$

Hence,

$$AB \neq BA.$$

Algebraic Properties of Matrix Multiplication

For compatible matrices A, B, C and scalar λ :

① **Associativity**

$$(AB)C = A(BC)$$

② **Left distributivity**

$$A(B + C) = AB + AC$$

③ **Right distributivity**

$$(A + B)C = AC + BC$$

④ **Scalar compatibility**

$$\lambda(AB) = (\lambda A)B = A(\lambda B)$$

⑤ **Identity**

$$AI = IA = A$$

Non-Commutativity

Unlike ordinary scalar multiplication,

$$AB \neq BA \quad \text{in general.}$$

The reason is structural: matrix multiplication represents **composition of transformations**, and the order in which transformations are performed matters.

Interpretation

If A represents a rotation and B represents a scaling, then

$$AB$$

means “apply B first, then A ”, whereas

$$BA$$

means “apply A first, then B ”.

Therefore, changing the order can produce a different geometric result.

Identity and Inverse Matrices

The identity matrix acts as the multiplicative identity:

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

For every $A \in \mathbb{R}^{n \times n}$,

$$AI_n = I_n A = A.$$

If A is invertible, then there exists A^{-1} such that

$$\boxed{AA^{-1} = A^{-1}A = I.}$$

A matrix is invertible precisely when

$$\boxed{\det(A) \neq 0.}$$

Thus matrix multiplication provides the algebraic framework for reversing an invertible linear transformation.

Matrix Multiplication and Linear Maps

Let

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

be a linear transformation. Relative to chosen bases, there exists a matrix

$$A \in \mathbb{R}^{m \times n}$$

such that

$$\boxed{T(\mathbf{x}) = A\mathbf{x}}.$$

If

$$S : \mathbb{R}^m \rightarrow \mathbb{R}^p$$

has matrix representation B , then

$$S(\mathbf{y}) = B\mathbf{y}.$$

Applying T followed by S gives

$$S(T(\mathbf{x})) = B(A\mathbf{x}) = (BA)\mathbf{x}.$$

Hence,

$$\boxed{[S \circ T] = [S][T]}$$

7. Matrix Multiplication as Composition

Suppose

$$T : V \rightarrow W, \quad S : W \rightarrow U$$

are linear transformations.

Their composition is

$$S \circ T : V \rightarrow U.$$

For every $\mathbf{x} \in V$,

$$(S \circ T)(\mathbf{x}) = S(T(\mathbf{x})).$$

If

$$[T] = A, \quad [S] = B,$$

then

$$[S \circ T] = BA.$$

Important

The **rightmost matrix acts first**.

$$BA\mathbf{x} = B(A\mathbf{x})$$

Why Does the Rightmost Matrix Act First?

Consider

$$\mathbf{x} \xrightarrow{A} A\mathbf{x} \xrightarrow{B} B(A\mathbf{x}).$$

Algebraically,

$$B(A\mathbf{x}) = (BA)\mathbf{x}.$$

Therefore,

$$BA = \text{"first } A, \text{ then } B"$$

$$\mathbf{x} \longrightarrow A\mathbf{x} \longrightarrow BA\mathbf{x}$$

This convention is fundamental in:

- robotics,
- computer graphics,
- computer vision,
- coordinate transformations,
- control systems.

Geometric Example: Rotation and Scaling

Consider the rotation matrix

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

and scaling matrix

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Applying scaling followed by rotation gives

$$R(\theta)S.$$

Applying rotation followed by scaling gives

$$SR(\theta).$$

In general,

$$\boxed{R(\theta)S \neq SR(\theta)}.$$

Thus the same two transformations can produce different final positions depending on their order.

A Concrete Transformation Example

Let

$$\theta = 90^\circ, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The rotation matrix is

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Let

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Scaling first:

$$S\mathbf{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad R(S\mathbf{x}) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Rotation first:

$$R\mathbf{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad S(R\mathbf{x}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Hence,

$$\boxed{R(S\mathbf{x}) \neq S(R\mathbf{x})}.$$

Composition Explains Associativity

Let

$$T_1 : V \rightarrow W, \quad T_2 : W \rightarrow U, \quad T_3 : U \rightarrow Z.$$

Composition satisfies

$$T_3 \circ (T_2 \circ T_1) = (T_3 \circ T_2) \circ T_1.$$

If their matrices are

$$A, \quad B, \quad C,$$

then

$$C(BA) = (CB)A.$$

Therefore, associativity of matrix multiplication is not merely an algebraic coincidence.

Structural meaning

Associativity follows from the associativity of function composition.

Generalisation to n -Dimensional Spaces

The same principle holds for

$$T_i : \mathbb{R}^{n_i} \rightarrow \mathbb{R}^{n_{i+1}}.$$

A sequence of transformations

$$T_k \circ T_{k-1} \circ \cdots \circ T_2 \circ T_1$$

is represented by

$$\boxed{A_k A_{k-1} \cdots A_2 A_1}.$$

Hence,

$$\mathbf{x}_{\text{final}} = A_k A_{k-1} \cdots A_1 \mathbf{x}_{\text{initial}}.$$

Interpretation

Matrix multiplication provides a compact algebraic representation of an entire chain of linear operations.

Practical Application: Robotics

In robotics, the position of a point attached to a robot arm undergoes a sequence of transformations.

For example,

$$\mathbf{x}_0 \xrightarrow{A_1} \mathbf{x}_1 \xrightarrow{A_2} \mathbf{x}_2 \xrightarrow{A_3} \mathbf{x}_3.$$

Therefore,

$$\boxed{\mathbf{x}_3 = A_3 A_2 A_1 \mathbf{x}_0}.$$

This is the mathematical foundation of **forward kinematics**.

- A_1 : transformation produced by joint 1.
- A_2 : transformation produced by joint 2.
- A_3 : transformation produced by joint 3.

The order cannot be arbitrarily changed because

$$A_3 A_2 A_1 \neq A_1 A_2 A_3$$

in general.

Applications in Computer Graphics

A point in a 2D image can undergo several transformations:

Object $\rightarrow$ Scale $\rightarrow$ Rotate $\rightarrow$ Translate $\rightarrow$ Screen.

In homogeneous coordinates, these operations can be represented using matrices.

A transformation pipeline can therefore be written as

$$\mathbf{x}_{screen} = T_{translate} R_{rotate} S_{scale} \mathbf{x}_{object}.$$

The rightmost operation acts first.

Key idea

Matrix multiplication converts a sequence of geometric operations into one composite transformation.

Applications in Computer Vision

In computer vision, transformations relate coordinate systems.

If

$$\mathbf{x}_C = R\mathbf{x}_W + \mathbf{t},$$

then R describes rotation and $\mathbf{t}$ describes translation.

For multiple coordinate transformations,

$$\mathbf{x}_3 = A_3 A_2 A_1 \mathbf{x}_0.$$

This is used in:

- camera calibration,
- 3D reconstruction,
- object tracking,
- image registration,
- simultaneous localisation and mapping (SLAM).

Key Takeaways

- 1 Matrix multiplication is defined through row-column inner products.

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}.$$

- 2 Matrix multiplication is associative and distributive:

$$(AB)C = A(BC).$$

- 3 Matrix multiplication is generally non-commutative:

$$AB \neq BA.$$

- 4 The deeper reason is that matrices represent linear maps, and matrix multiplication represents their composition.

- 5 In

$$BA\mathbf{x},$$

the rightmost matrix A acts first.

- 6 This framework generalises naturally to transformation chains in robotics, graphics and computer vision.

8. Linear Transformation

Definition

Let V and W be vector spaces over a field $\mathbb{F}$. A map

$$T : V \rightarrow W$$

is called a **linear transformation** if

$$T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$$

for all $\mathbf{u}, \mathbf{v} \in V$ and $\alpha, \beta \in \mathbb{F}$.

Equivalently, linearity consists of:

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$$

and

$$T(\alpha \mathbf{u}) = \alpha T(\mathbf{u}).$$

Fundamental Properties

Every linear transformation satisfies

$$T(\mathbf{0}) = \mathbf{0}.$$

If

$$\mathbf{u} = \sum_{i=1}^n c_i \mathbf{v}_i,$$

then

$$T(\mathbf{u}) = \sum_{i=1}^n c_i T(\mathbf{v}_i).$$

Therefore, linear transformations preserve linear combinations.

Key principle

A linear transformation does not need to be evaluated separately on every vector. Its action on a basis completely determines its action on the entire vector space.

A Linear Transformation is Determined by a Basis

Let

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

be a basis of V .

Every vector $\mathbf{x} \in V$ has a unique representation

$$\mathbf{x} = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n.$$

By linearity,

$$T(\mathbf{x}) = c_1 T(\mathbf{v}_1) + \dots + c_n T(\mathbf{v}_n).$$

Hence,

T is completely determined by $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$.

This is the fundamental reason matrices can represent linear transformations.

Matrix Representation of a Linear Transformation

Let

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

With respect to chosen bases, there exists a unique matrix

$$A \in \mathbb{R}^{m \times n}$$

such that

$$T(\mathbf{x}) = A\mathbf{x}.$$

If

$$\mathbf{e}_1, \dots, \mathbf{e}_n$$

are the standard basis vectors, then

$$A = \left[\begin{array}{c|c|c|c} & & & \\ T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \\ & & & \end{array} \right].$$

Thus the columns of A are the images of the basis vectors.

Worked Example: Constructing a Matrix

Consider

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

defined by

$$T(x, y) = (2x + y, x - 3y).$$

For the standard basis,

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$T(\mathbf{e}_1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 1 \\ -3 \end{bmatrix}.$$

Hence the matrix representation is

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}.$$

Kernel and Image

Let

$$T : V \rightarrow W$$

be linear.

Kernel

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$$

The kernel contains all vectors that are mapped to the zero vector.

Image

$$\operatorname{Im}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\}$$

The image is the set of all vectors that can actually be produced by the transformation.

Geometric Meaning of Kernel and Image

For

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

- $\ker(T)$ describes directions that are completely collapsed by the transformation.
- $\text{Im}(T)$ describes the accessible output directions.

For a matrix A ,

$$\ker(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$$

and

$$\text{Im}(A) = \text{Col}(A).$$

Therefore,

Kernel $\longleftrightarrow$ information lost,

while

Image $\longleftrightarrow$ information retained in the output space.

Rank and Nullity

The **rank** of T is

$$\text{rank}(T) = \dim(\text{Im}(T)).$$

The **nullity** is

$$\text{nullity}(T) = \dim(\ker(T)).$$

For a matrix A ,

$$\text{rank}(A) = \dim(\text{Col}(A)).$$

Rank measures the number of independent output directions, while nullity measures the number of independent directions that are mapped to zero.

Rank–Nullity Theorem

Let

$$T : V \rightarrow W$$

be a linear transformation where V is finite-dimensional.

Then

$$\dim(V) = \text{rank}(T) + \text{nullity}(T).$$

For a matrix

$$A \in \mathbb{R}^{m \times n},$$

$$n = \text{rank}(A) + \text{nullity}(A).$$

Interpretation

The number of input degrees of freedom is divided between independent output directions and directions lost in the kernel.

Practical Interpretation: Data Science

Suppose a dataset contains n features represented by vectors

$$\mathbf{x}_1, \dots, \mathbf{x}_n.$$

If the transformation matrix has

$$\text{rank}(A) < n,$$

then the transformation cannot preserve all independent directions in the original feature space.

Consequently,

$$\text{nullity}(A) > 0$$

means that some non-zero directions are mapped to zero.

Interpretation

Rank reveals the effective dimensionality of the information represented after a linear transformation.

Linear Transformations: General Structure

For

$$T : V \rightarrow W,$$

the following structure is fundamental:

$$\boxed{V \xrightarrow{T} W}$$

with

$$\ker(T) \subseteq V$$

and

$$\operatorname{Im}(T) \subseteq W.$$

The dimensions satisfy

$$\boxed{\dim V = \dim \ker(T) + \dim \operatorname{Im}(T).}$$

This framework applies to finite-dimensional vector spaces over $\mathbb{R}$, $\mathbb{C}$, and more general fields.

9. Three-Dimensional Linear Transformations

A linear transformation in three-dimensional space is a map

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

represented by

$$\mathbf{x}' = A\mathbf{x}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

Depending on A , the transformation can perform:

- scaling,
- rotation,
- reflection,
- projection,
- shearing.

3D Scaling

A general scaling transformation is

$$S = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Hence,

$$(x', y', z') = (s_x x, s_y y, s_z z).$$

If

$$s_x = s_y = s_z,$$

the scaling is uniform.

If the factors differ, the transformation is anisotropic.

3D Rotation About the z-Axis

Rotation through angle θ about the z-axis is represented by

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus,

$$\mathbf{x}' = R_z(\theta)\mathbf{x}.$$

The z-coordinate remains unchanged while the (x, y) coordinates rotate in the xy -plane.

Similar matrices describe rotations about the x and y axes.

3D Reflection

Reflection across the xy -plane is represented by

$$R_{xy} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

Therefore,

$$(x, y, z) \mapsto (x, y, -z).$$

Similarly,

$$R_{yz} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

represents reflection across the yz -plane.

3D Shearing

A shear can alter one coordinate according to another.
For example, shearing x with respect to z :

$$H = \begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x + kz \\ y \\ z \end{bmatrix}.$$

The transformation changes shape while preserving certain parallel relationships.

Orthogonal Transformations

A matrix $Q \in \mathbb{R}^{n \times n}$ is orthogonal if

$$Q^T Q = Q Q^T = I.$$

Consequently,

$$Q^{-1} = Q^T.$$

For any vectors $\mathbf{x}, \mathbf{y}$,

$$(Q\mathbf{x})^T (Q\mathbf{y}) = \mathbf{x}^T Q^T Q \mathbf{y} = \mathbf{x}^T \mathbf{y}.$$

Hence orthogonal transformations preserve inner products.

Why Orthogonal Matrices Preserve Lengths and Angles

Since

$$Q^T Q = I,$$

the norm satisfies

$$\|Q\mathbf{x}\|^2 = (Q\mathbf{x})^T (Q\mathbf{x}) = \mathbf{x}^T Q^T Q \mathbf{x} = \mathbf{x}^T \mathbf{x}.$$

Therefore,

$$\boxed{\|Q\mathbf{x}\| = \|\mathbf{x}\|}.$$

Similarly,

$$(Q\mathbf{x})^T (Q\mathbf{y}) = \mathbf{x}^T \mathbf{y}.$$

Thus angles are preserved.

Geometric consequence

Orthogonal transformations preserve distances, lengths and angles. They represent rigid geometric motions such as rotations and reflections.

Rotation Matrices and Determinant

A proper 3D rotation matrix satisfies

$$R^T R = I$$

and

$$\det(R) = +1.$$

Reflections also satisfy orthogonality but have

$$\det(Q) = -1.$$

Therefore,

$$Q \in O(3)$$

represents an orthogonal transformation, while

$$R \in SO(3)$$

represents an orientation-preserving rotation.

Composition of 3D Transformations

Suppose

$$\mathbf{x}_1 = A_1 \mathbf{x}_0$$

followed by

$$\mathbf{x}_2 = A_2 \mathbf{x}_1.$$

Then

$$\mathbf{x}_2 = A_2 A_1 \mathbf{x}_0.$$

For a sequence of transformations,

$$\mathbf{x}_n = A_n A_{n-1} \cdots A_2 A_1 \mathbf{x}_0.$$

The rightmost matrix acts first.

This is the fundamental mathematical structure behind transformation pipelines in 3D systems.

Application: Robotics

A robot manipulator consists of multiple links and joints.
Each joint contributes a transformation

$$A_1, A_2, \dots, A_n.$$

The end-effector position is obtained by

$$\mathbf{x}_{EE} = A_n A_{n-1} \cdots A_1 \mathbf{x}_0.$$

Therefore, matrix multiplication provides the mathematical foundation of **forward kinematics**.

- Joint transformations describe local motion.
- Their matrix product describes global motion.
- Changing the order changes the physical configuration.

Application: Computer Graphics and CAD

A 3D object can undergo a sequence of transformations:

Object $\rightarrow$ Scale $\rightarrow$ Rotate $\rightarrow$ Translate.

Mathematically,

$$\mathbf{x}_{final} = T_{\text{translation}} R_{\text{rotation}} S_{\text{scale}} \mathbf{x}_{initial}.$$

In practical graphics systems, homogeneous coordinates extend this framework to include translation in the same matrix-based transformation pipeline.

Applications include:

- CAD modelling,
- animation,
- 3D rendering,
- virtual reality,
- computer-aided engineering.

Application: Computer Vision

In computer vision, different coordinate frames are related through transformations.

A rigid transformation can be expressed as

$$\mathbf{x}' = R\mathbf{x} + \mathbf{t}$$

where

$$R \in SO(3)$$

represents rotation and

$$\mathbf{t} \in \mathbb{R}^3$$

represents translation.

This framework is essential for:

- camera pose estimation,
- 3D reconstruction,
- object tracking,
- SLAM.

Summary: Linear Transformations

$$T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$$

A linear transformation is completely determined by its action on a basis.

$$T(\mathbf{x}) = A\mathbf{x}$$

The central associated spaces are

$$\ker(T), \quad \operatorname{Im}(T)$$

with

$$\dim V = \operatorname{rank}(T) + \operatorname{nullity}(T).$$

Summary: 3D Transformations

Three-dimensional linear transformations can represent

Scaling, Rotation, Reflection, Projection, Shearing

Orthogonal transformations satisfy

$$Q^T Q = I$$

and preserve

lengths, distances, angles and inner products.

A chain of transformations is represented by

$$A_n A_{n-1} \cdots A_1.$$

This mathematical framework underlies modern

Robotics + CAD + Computer Graphics + Computer Vision.

10. Determinant

Definition

The determinant is a scalar-valued function

$$\det : M_n(\mathbb{F}) \rightarrow \mathbb{F}$$

that associates a scalar with every square matrix.

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

The determinant provides information about:

- volume scaling,
- orientation,
- linear independence,
- invertibility

Geometric Meaning of the Determinant

Let

$$T(\mathbf{x}) = A\mathbf{x}$$

be a linear transformation on $\mathbb{R}^n$.

Then

$$|\det(A)| = \text{volume scaling factor.}$$

Therefore, if a region has volume V , then after applying A ,

$$V_{\text{new}} = |\det(A)| V_{\text{old}}.$$

In different dimensions:

Dimension	Geometric quantity
1	length
2	area
3	volume
n	n -dimensional volume

Sign of the Determinant and Orientation

The magnitude

$$|\det(A)|$$

describes the amount of volume scaling.

The sign describes orientation:

$$\det(A) > 0 \Rightarrow \text{orientation preserved}$$

$$\det(A) < 0 \Rightarrow \text{orientation reversed}$$

$$\det(A) = 0 \Rightarrow \text{dimensional collapse}$$

For example, a reflection reverses orientation and therefore has a negative determinant.

Determinant of a 2×2 Matrix

Consider

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

The determinant can be interpreted geometrically as the signed area of the parallelogram generated by the columns of A .

If

$$\mathbf{u} = \begin{bmatrix} a \\ c \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} b \\ d \end{bmatrix},$$

then

$$\det(A) = \mathbf{u}_1 \mathbf{v}_2 - \mathbf{u}_2 \mathbf{v}_1.$$

Determinant of a 3×3 Matrix

For

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

the determinant is

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Equivalently,

$$\det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

This is the expansion by cofactors along the first row.

Cofactor Expansion

For a matrix $A = [a_{ij}]$, define the minor

$$M_{ij} = \det(A_{ij}),$$

where A_{ij} is obtained by deleting row i and column j .

The cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The determinant can be expanded along any row:

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij}.$$

Or along any column:

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij}.$$

Important Properties of Determinants

For square matrices A, B :

1

$$\det(AB) = \det(A) \det(B)$$

2

$$\det(A^T) = \det(A)$$

3 If A is invertible,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

4 For scalar λ ,

$$\det(\lambda A) = \lambda^n \det(A)$$

for $A \in M_n(\mathbb{F})$.

5

$$\det(I) = 1.$$

Elementary Row Operations

Determinants behave predictably under elementary row operations.

- 1 Interchanging two rows changes the sign:

$$\det(A') = -\det(A)$$

- 2 Multiplying a row by k multiplies the determinant by k :

$$\det(A') = k \det(A)$$

- 3 Adding a multiple of one row to another leaves the determinant unchanged:

$$\det(A') = \det(A).$$

These properties make Gaussian elimination an efficient method for determinant computation.

Determinant and Linear Independence

Let

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n] .$$

Then

$$\det(A) = 0 \iff \{\mathbf{a}_1, \dots, \mathbf{a}_n\} \text{ is linearly dependent.}$$

Equivalently,

$$\det(A) \neq 0 \iff \text{columns of } A \text{ are linearly independent.}$$

Thus the determinant provides an algebraic test for whether a set of vectors forms a basis of $\mathbb{R}^n$.

Determinant and Invertibility

For

$$A \in M_n(\mathbb{F}),$$

the following statements are equivalent:

$$\boxed{\det(A) \neq 0}$$


$$\boxed{A \text{ is invertible}}$$


$$\boxed{\ker(A) = \{\mathbf{0}\}}$$



$$\boxed{\text{rank}(A) = n}$$

This is one of the central equivalence results in linear algebra.

Abstract Definition of the Determinant

At an abstract level, the determinant is the unique function

$$\det : M_n(\mathbb{F}) \rightarrow \mathbb{F}$$

satisfying:

- ① **Multilinearity** in the rows,
- ② **Alternating property**: if two rows are equal,

$$\det(A) = 0,$$

- ③ **Normalization**:

$$\det(I_n) = 1.$$

These properties completely characterize the determinant.

Conceptual interpretation

The determinant is not merely a formula. It is a multilinear alternating measure of oriented n -dimensional volume.

Practical Example: Volume Scaling

Consider

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

Then

$$\det(A) = 2(3)(4) = 24.$$

Therefore, the transformation

$$\mathbf{x}' = A\mathbf{x}$$

multiplies every three-dimensional volume by

$$\boxed{24}.$$

For a unit cube,

$$V_{\text{original}} = 1,$$

and therefore

$$V_{\text{transformed}} = 24.$$

11. Inverse Matrix

Let

$$A \in M_n(\mathbb{F}).$$

The matrix A^{-1} is called the inverse of A if

$$AA^{-1} = A^{-1}A = I_n.$$

A matrix possessing an inverse is called **invertible** or **nonsingular**.
If no inverse exists, the matrix is called **singular**.

Existence condition

$$A^{-1} \text{ exists} \iff \det(A) \neq 0.$$

Inverse as an Inverse Linear Transformation

Let

$$T(\mathbf{x}) = A\mathbf{x}.$$

If A is invertible, then

$$\mathbf{x}' = A\mathbf{x}$$

can be reversed:

$$\mathbf{x} = A^{-1}\mathbf{x}'.$$

Therefore,

$$T^{-1}(\mathbf{x}) = A^{-1}\mathbf{x}.$$

The inverse transformation satisfies

$$T^{-1} \circ T = I$$

and

$$T \circ T^{-1} = I.$$

Hence matrix inversion is the coordinate representation of reversing an invertible linear transformation.

Inverse of a 2×2 Matrix

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

if

$$ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Equivalently,

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}.$$

Thus the determinant appears directly in the inverse formula.

Adjugate Formula

For a nonsingular matrix A ,

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

The adjugate is the transpose of the cofactor matrix:

$$\operatorname{adj}(A) = [C_{ij}]^T.$$

The fundamental identity is

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = \det(A)I.$$

Therefore, when $\det(A) \neq 0$, division by the determinant produces the inverse.

Inverse and Systems of Linear Equations

Consider

$$A\mathbf{x} = \mathbf{b}.$$

If A is invertible, multiply both sides by A^{-1} :

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}.$$

Hence,

$$\boxed{\mathbf{x} = A^{-1}\mathbf{b}.$$

Therefore, an invertible coefficient matrix guarantees a unique solution for every $\mathbf{b}$.

Important

Matrix inversion converts a system of simultaneous equations into a direct linear transformation of the right-hand side.

Worked Example: Solving a Linear System

Consider

$$\begin{cases} 2x + y = 5, \\ x + 3y = 6. \end{cases}$$

Write

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Since

$$\det(A) = 6 - 1 = 5 \neq 0,$$

A is invertible.

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}.$$

Hence,

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{5} \begin{bmatrix} 9 \\ 7 \end{bmatrix}.$$

Therefore,

$$\boxed{x = \frac{9}{5}, \quad y = \frac{7}{5}.}$$

Inverse Matrix: Important Properties

For invertible matrices A and B :

1

$$(AB)^{-1} = B^{-1}A^{-1}$$

2

$$(A^T)^{-1} = (A^{-1})^T$$

3

$$(A^{-1})^{-1} = A$$

4

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

5

$$(A^k)^{-1} = (A^{-1})^k$$

for integer k .

Why the Order Reverses in $(AB)^{-1}$

We want

$$(AB)(AB)^{-1} = I.$$

Consider

$$(AB)(B^{-1}A^{-1}).$$

By associativity,

$$A(BB^{-1})A^{-1} = AIA^{-1} = I.$$

Therefore,

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Interpretation

To reverse a sequence of transformations, the operations must be undone in reverse order.

Inverse Transformations in Coordinate Systems

Suppose a coordinate transformation is

$$\mathbf{x}' = A\mathbf{x}.$$

To recover the original coordinates,

$$\mathbf{x} = A^{-1}\mathbf{x}'.$$

Thus:

Forward transformation: A

Reverse transformation: A^{-1}

This principle is widely used in:

- robotics,
- computer graphics,
- computer vision,
- coordinate geometry,
- control systems

Singular Matrix and Information Loss

If

$$\det(A) = 0,$$

then

A^{-1} does not exist.

Geometrically, the transformation collapses an n -dimensional region into a lower-dimensional set.

For example,

$$\mathbb{R}^3 \longrightarrow \text{a plane}$$

or

$$\mathbb{R}^2 \longrightarrow \text{a line.}$$

Once information is lost through this dimensional collapse, the original vector cannot generally be reconstructed.

Determinant–Inverse–Rank Connection

For

$$A \in M_n(\mathbb{F}),$$

the following are equivalent:

$$\det(A) \neq 0$$



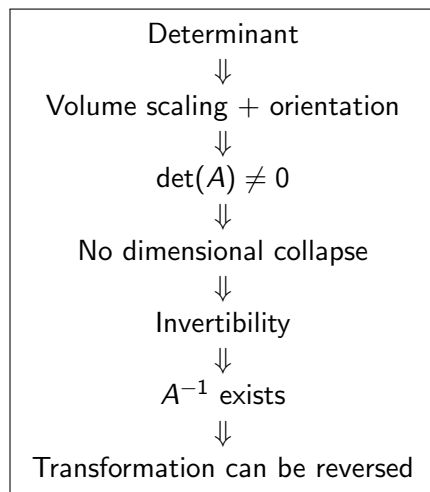
$$A^{-1} \text{ exists}$$



$$\text{rank}(A) = n$$



Final Conceptual Picture



Therefore determinant and inverse are not isolated computational concepts. They describe fundamental geometric and algebraic properties of linear transformations.

12. Nonsquare Matrices and Linear Transformations

A matrix need not be square.

Let

$$A \in \mathbb{R}^{m \times n}.$$

Then A defines a linear transformation

$$T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^m, \quad T_A(x) = Ax.$$

Interpretation

- n = dimension of the domain
- m = dimension of the codomain
- A has n columns and m rows
- The transformation may change dimension.

Thus,

$$A_{m \times n} : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

For example,

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Rank and Nullity of a Nonsquare Matrix

For

$$A \in \mathbb{R}^{m \times n},$$

the **rank** is

$$\text{rank}(A) = \dim(\text{Col}(A))$$

and the **nullity** is

$$\text{nullity}(A) = \dim(\ker A).$$

The Rank–Nullity Theorem states

$$\text{rank}(A) + \text{nullity}(A) = n$$

where n is the number of columns of A .

Important

Rank measures the number of independent output directions.

Nullity measures the number of independent directions that are lost.

Example: A 2×3 Matrix

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

The columns are

$$a_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Since

$$a_2 = 2a_1, \quad a_3 = 3a_1,$$

there is only one independent column direction.

Therefore,

$$\boxed{\text{rank}(A) = 1}.$$

Since A has three columns,

Why Nonsquare Matrices Usually Have No Ordinary Inverse

For a square matrix

$$A \in \mathbb{R}^{n \times n},$$

an inverse may satisfy

$$AA^{-1} = A^{-1}A = I_n.$$

For a nonsquare matrix

$$A \in \mathbb{R}^{m \times n}, \quad m \neq n,$$

an ordinary two-sided inverse cannot exist.

However, one-sided inverses may exist.

Full column rank

If

$$A \in \mathbb{R}^{m \times n}, \quad m \geq n, \quad \text{rank}(A) = n,$$

then A has a left inverse L satisfying

Moore–Penrose Pseudoinverse

For a general matrix A , the **Moore–Penrose pseudoinverse**

$$A^+$$

plays the role of a generalized inverse.

For a full-column-rank matrix,

$$A^+ = (A^T A)^{-1} A^T.$$

It is particularly important for overdetermined systems

$$Ax = b, \quad m > n.$$

The least-squares solution is

$$\hat{x} = A^+ b.$$

When A has full column rank,

$$\hat{x} = (A^T A)^{-1} A^T b.$$

Geometric Interpretation of a Nonsquare Transformation

A transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

can be represented by a 2×3 matrix.

It maps a three-dimensional domain into a two-dimensional codomain.

$$\mathbb{R}^3 \xrightarrow{T} \mathbb{R}^2.$$

The transformation may therefore:

- compress dimensions,
- eliminate redundant directions,
- project data onto a lower-dimensional space,
- preserve only the directions represented in the image.

If

$$\text{rank}(A) < 3,$$

then some nonzero directions in $\mathbb{R}^3$ are mapped to zero.

These directions form the null space.

13. Column Space

Let

$$A = \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

The **column space** is

$$\text{Col}(A) = \text{span}\{a_1, a_2, \dots, a_n\}.$$

It consists of all possible linear combinations of the columns.
Since

$$Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n,$$

we obtain

$$\text{Col}(A) = \{Ax : x \in \mathbb{R}^n\}.$$

Therefore,

Column Space and Consistency

Consider the linear system

$$Ax = b.$$

The vector b belongs to the column space exactly when the system is consistent.

$$Ax = b \text{ is consistent} \iff b \in \text{Col}(A).$$

Indeed,

$$Ax = b$$

means

$$x_1 a_1 + \cdots + x_n a_n = b.$$

Thus b must be a linear combination of the columns of A .

Geometric meaning

The column space is precisely the set of all outputs that the transformation $x \mapsto Ax$ can produce.

If

Example: Column Space

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

Since

$$a_2 = 2a_1, \quad a_3 = 3a_1,$$

we obtain

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$$

Hence

$$\text{Col}(A) = \left\{ \begin{bmatrix} t \\ 2t \end{bmatrix} : t \in \mathbb{R} \right\}.$$

Therefore, the system

Rank as the Dimension of Column Space

The rank of A is

$$\text{rank}(A) = \dim(\text{Col}(A)).$$

Thus rank counts the number of independent output directions.

If

$$A \in \mathbb{R}^{m \times n},$$

then

$$\text{rank}(A) \leq \min(m, n).$$

Special cases:

$$\text{rank}(A) = n$$

means all columns are linearly independent.

Similarly,

Column Space in Data Science

Suppose a data matrix is

$$X = \begin{bmatrix} | & | & \cdots & | \\ x_1 & x_2 & \cdots & x_p \\ | & | & \cdots & | \end{bmatrix}.$$

Each column can represent a feature.

Then

$$\text{Col}(X)$$

contains all vectors that can be generated as linear combinations of the available features.

If one feature satisfies

$$x_3 = 2x_1 - x_2,$$

then x_3 adds no new column-space direction.

Therefore,

14. Null Space / Kernel

For

$$A \in \mathbb{R}^{m \times n},$$

the **null space** is

$$\text{Null}(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

It is also called the kernel of the associated linear transformation:

$$\text{Null}(A) = \ker(T_A).$$

Thus,

$$x \in \text{Null}(A)$$

means that x is mapped to the zero vector.

Geometric meaning

The null space contains all directions in the domain that are completely eliminated by the transformation.

Example: Finding the Null Space

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

We solve

$$Ax = 0.$$

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Then

$$x_1 + 2x_2 + 3x_3 = 0.$$

Hence

Null Space and Information Loss

Suppose

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

If

$$u \in \ker(T), \quad u \neq 0,$$

then

$$T(u) = 0.$$

Moreover, for any x ,

$$T(x + u) = T(x) + T(u) = T(x).$$

Therefore,

$$T(x + u) = T(x)$$

whenever $u \in \ker(T)$.

Rank–Nullity Theorem

Let

$$T : V \rightarrow W$$

be a linear transformation with finite-dimensional domain V .
Then

$$\dim(V) = \dim(\ker T) + \dim(\operatorname{Im} T).$$

For a matrix

$$A \in \mathbb{R}^{m \times n},$$

this becomes

$$n = \operatorname{nullity}(A) + \operatorname{rank}(A).$$

Equivalently,

$$\operatorname{nullity}(A) = n - \operatorname{rank}(A).$$

Geometric Interpretation of Rank–Nullity

For

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2,$$

suppose

$$\text{rank}(T) = 2.$$

Then

$$\text{nullity}(T) = 3 - 2 = 1.$$

So:

$\dim(\text{Im } T) = 2,$	$\dim(\ker T) = 1.$
---------------------------	---------------------

Geometrically:

- The image is a plane in $\mathbb{R}^2$ (possibly all of $\mathbb{R}^2$).
- One independent direction in $\mathbb{R}^3$ is collapsed to zero.

Null Space and Homogeneous Systems

The homogeneous system

$$Ax = 0$$

always has the trivial solution

$$x = 0.$$

The complete solution set is

$$\boxed{\text{Null}(A)}.$$

If

$$\text{Null}(A) = \{0\},$$

then the homogeneous system has only the trivial solution.
This implies that the columns of A are linearly independent.
For a square matrix,

$$\boxed{\text{Null}(A) = \{0\} \iff A \text{ is invertible.}}$$

Rank, Nullity and Invertibility

For

$$A \in \mathbb{R}^{n \times n},$$

the following statements are equivalent:

$$\begin{aligned} A \text{ is invertible} &\iff \det(A) \neq 0 \\ &\iff \text{rank}(A) = n \\ &\iff \text{Null}(A) = \{0\} \\ &\iff \text{columns of } A \text{ are linearly independent} \\ &\iff Ax = 0 \text{ has only } x = 0. \end{aligned}$$

This is one of the central equivalence chains of linear algebra.

Conceptual chain

$$\text{No information lost} \iff \ker A = \{0\} \iff \text{rank}(A) = n \iff A^{-1} \text{ exists}$$

Applications: Least Squares

For an overdetermined system

$$Ax \approx b, \quad A \in \mathbb{R}^{m \times n}, \quad m > n,$$

an exact solution may not exist. We instead minimize

$$\|Ax - b\|_2^2.$$

The optimal solution satisfies the normal equations

$$A^T A \hat{x} = A^T b.$$

If A has full column rank,

$$\hat{x} = (A^T A)^{-1} A^T b = A^+ b.$$

The vector

$$A\hat{x}$$

is the orthogonal projection of b onto $\text{Col}(A)$.

Thus the geometry of column space directly explains least-squares regression.

Applications in Machine Learning

Suppose

$$X \in \mathbb{R}^{m \times p}$$

is a data matrix.

- $\text{Col}(X)$ represents the space generated by the features.
- $\text{rank}(X)$ measures independent feature directions.
- $\text{Null}(X)$ describes parameter directions invisible to the data.

If

$$Xv = 0,$$

then changing model parameters in the direction v produces no change in the predicted output generated by X .

Hence a nontrivial null space can indicate

parameter non-identifiability.

This is important in:

- linear regression.

15. Dot Product

For vectors

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n,$$

the **dot product** is defined by

$$u \cdot v = u^T v = \sum_{i=1}^n u_i v_i.$$

The result is a scalar.

For example,

$$u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix},$$

gives

Geometric Meaning of the Dot Product

For nonzero vectors u, v ,

$$u \cdot v = \|u\| \|v\| \cos \theta$$

where θ is the angle between them.

Therefore,

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}.$$

This gives the three important cases:

$$u \cdot v > 0 \implies \theta < 90^\circ,$$

$$u \cdot v = 0 \implies \theta = 90^\circ,$$

$$u \cdot v < 0 \implies \theta > 90^\circ.$$

In particular,

$$u \cdot v = 0 \iff u \perp v$$

Properties of the Dot Product

For $u, v, w \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$:

① Symmetry

$$u \cdot v = v \cdot u.$$

② Linearity

$$(\alpha u + \beta v) \cdot w = \alpha(u \cdot w) + \beta(v \cdot w).$$

③ Positive definiteness

$$u \cdot u \geq 0, \quad u \cdot u = 0 \iff u = 0.$$

④ Norm induced by the dot product

$$\boxed{\|u\| = \sqrt{u \cdot u}}.$$

These properties are more fundamental than the particular coordinate formula.

They motivate the abstract concept of an **inner product**.

Inner Product: Linear Algebra Generalization

An **inner product** on a real vector space V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

satisfying, for $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$,

$$\langle u, v \rangle = \langle v, u \rangle,$$

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle,$$

and

$$\langle u, u \rangle \geq 0, \quad \langle u, u \rangle = 0 \iff u = 0.$$

The dot product is therefore one particular inner product:

$$\langle u, v \rangle = u^T v.$$

Norm and Distance from an Inner Product

An inner product induces a norm:

$$\|u\| = \sqrt{\langle u, u \rangle}.$$

It also induces a distance:

$$d(u, v) = \|u - v\|.$$

The angle between two nonzero vectors is defined by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

Orthogonality is defined abstractly as

$$u \perp v \iff \langle u, v \rangle = 0.$$

Thus the inner product provides the complete geometric structure needed for Euclidean-style reasoning in an abstract vector space.

Orthogonal Projection

Let $u \neq 0$.

The projection of v onto u is

$$\text{proj}_u(v) = \frac{\langle v, u \rangle}{\langle u, u \rangle} u.$$

For the Euclidean dot product,

$$\text{proj}_u(v) = \frac{v \cdot u}{u \cdot u} u.$$

The residual

$$r = v - \text{proj}_u(v)$$

satisfies

$$r \perp u.$$

Hence

$$v = \underbrace{\text{proj}_u(v)}_{\text{parallel component}} + \underbrace{r}_{\text{orthogonal component}}.$$

This decomposition is fundamental in least squares, Fourier analysis and

16. Function Spaces as Vector Spaces

A set of functions can itself form a vector space.

For example,

$$V = C[a, b]$$

is the vector space of continuous real-valued functions on $[a, b]$.

For

$$f, g \in V, \quad \alpha, \beta \in \mathbb{R},$$

define

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x).$$

Thus functions can be added and multiplied by scalars exactly like vectors.

This allows us to extend linear algebra from

$$\mathbb{R}^n$$

to infinite-dimensional spaces.

Inner Product on a Function Space

For suitable real-valued functions f, g on $[a, b]$, define

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx.$$

This is an inner product on an appropriate function space.
The corresponding norm is

$$\|f\| = \sqrt{\int_a^b f(x)^2 dx}.$$

Two functions are orthogonal if

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx = 0.$$

Therefore, orthogonality is not restricted to arrows in Euclidean space.
It can describe relationships between functions.

Example of Orthogonal Functions

Consider the functions

$$f(x) = 1, \quad g(x) = x$$

on the interval $[-1, 1]$.

Their inner product is

$$\langle f, g \rangle = \int_{-1}^1 x \, dx.$$

Since x is an odd function,

$$\int_{-1}^1 x \, dx = 0.$$

Hence

$$\boxed{\langle 1, x \rangle = 0}$$

and therefore

Orthogonal Bases of Functions

A collection

$$\{f_1, f_2, \dots, f_n\}$$

is orthogonal if

$$\langle f_i, f_j \rangle = 0, \quad i \neq j.$$

If every function f in a space can be represented as

$$f = \sum_i c_i f_i,$$

then $\{f_i\}$ acts as a basis.

For an orthogonal basis, coefficients can be obtained using

$$c_i = \frac{\langle f, f_i \rangle}{\langle f_i, f_i \rangle}.$$

This is directly analogous to expressing a vector in an orthogonal basis of

$\mathbb{R}^n$

Applications of Function-Space Inner Products

Inner products on function spaces provide the mathematical foundation for:

- **Fourier analysis**

$$f(x) \approx \sum_k c_k e^{ikx},$$

- **Orthogonal polynomials**

$$P_0(x), P_1(x), P_2(x), \dots$$

- **Approximation theory**
- **Signal processing**
- **Numerical methods**
- **Differential equations**

The central idea is

vectors $\longrightarrow$ functions

while retaining

inner product $\longrightarrow$ length + angle + orthogonality + projection.

Least-Squares Approximation in Function Spaces

Suppose we approximate a function $f(x)$ by

$$p(x) = c_1\phi_1(x) + \cdots + c_n\phi_n(x).$$

We minimize the squared error

$$\|f - p\|^2 = \int_a^b [f(x) - p(x)]^2 dx.$$

At the optimum, the error is orthogonal to every basis function:

$$\langle f - p, \phi_i \rangle = 0, \quad i = 1, \dots, n.$$

This produces the normal equations

$$\sum_{j=1}^n c_j \langle \phi_j, \phi_i \rangle = \langle f, \phi_i \rangle.$$

Thus least squares in function spaces is a direct generalization of ordinary linear algebra.

17. Cross Product in $\mathbb{R}^3$

For

$$u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3,$$

the cross product is

$$u \times v = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

It produces another vector in $\mathbb{R}^3$.

Its defining geometric property is

$$u \times v \perp u, \quad u \times v \perp v.$$

Thus the cross product constructs a vector normal to the plane spanned by u and v .

Magnitude of the Cross Product

The magnitude satisfies

$$\|u \times v\| = \|u\| \|v\| \sin \theta$$

where θ is the angle between u and v .

But

$$\|u\| \|v\| \sin \theta$$

is exactly the area of the parallelogram generated by u and v .

Therefore,

$$\text{Area of parallelogram} = \|u \times v\|.$$

The area of the corresponding triangle is

$$\frac{1}{2} \|u \times v\|.$$

If

Algebraic Properties of the Cross Product

For $u, v, w \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$:

$$u \times v = -(v \times u)$$

so the cross product is **anticommutative**.

Also,

$$u \times u = 0$$

and

$$u \times (v + w) = u \times v + u \times w.$$

Scalar compatibility gives

$$(\alpha u) \times v = \alpha(u \times v).$$

However,

$$u \times v \neq v \times u$$

Cross Product and Orientation

The cross product follows the right-hand rule:

$$e_1 \times e_2 = e_3, \quad e_2 \times e_3 = e_1, \quad e_3 \times e_1 = e_2.$$

Reversing the order reverses the direction:

$$e_2 \times e_1 = -e_3.$$

Therefore, the cross product contains both:

- magnitude information,
- orientation information.

This makes it especially useful in three-dimensional geometry and mechanics.

Cross Product vs. Inner Product

	Inner Product	Cross Product
Input	u, v	u, v
Output	Scalar	Vector
Standard space	$\mathbb{R}^n$	$\mathbb{R}^3$
Symmetry	$\langle u, v \rangle = \langle v, u \rangle$	$u \times v = -(v \times u)$
Measures	Angle/alignment	Area + normal
Orthogonality	$u \perp v$ if $\langle u, v \rangle = 0$	Output perpendicular to both

Important Generalisation

Inner products exist on many abstract vector spaces.

The standard vector cross product is a special construction associated with three-dimensional Euclidean geometry and does not generalize to arbitrary vector spaces.

Applications of the Cross Product

The cross product appears naturally in:

- **Torque**

$$\tau = r \times F$$

- **Angular momentum**

$$L = r \times p$$

- **Electromagnetism**
- **3D geometry**
- **Computer graphics**
- **Robotics**

For example, torque is perpendicular to both the position vector r and the applied force F .

Its magnitude is

$$\|\tau\| = \|r\| \|F\| \sin \theta.$$

Cross Product and Surface Normals

Suppose a surface contains two tangent vectors

$$u, \quad v.$$

A normal vector can be constructed as

$$n = u \times v.$$

The corresponding unit normal is

$$\hat{n} = \frac{u \times v}{\|u \times v\|}.$$

This is fundamental in computer graphics because surface normals determine:

- lighting,
- shading,
- visibility,
- reflection,
- orientation of surfaces.

Cross Product Under a Linear Transformation

Let

$$A \in GL(3, \mathbb{R})$$

be an invertible 3×3 matrix.

A fundamental identity is

$$(Au) \times (Av) = \det(A) A^{-T}(u \times v)$$

where

$$A^{-T} = (A^{-1})^T.$$

This is different from the ordinary transformation rule

$$Au.$$

The cross product behaves according to the determinant and the inverse transpose.

Why Does the Inverse Transpose Appear?

Let

$$n = u \times v.$$

Since n is perpendicular to both u and v ,

$$n^T u = 0, \quad n^T v = 0.$$

After transformation, the new normal n' must satisfy

$$n'^T (Au) = 0$$

and

$$n'^T (Av) = 0.$$

Therefore,

$$(A^T n')^T u = 0$$

and

Special Case: Proper Rotations

For a proper rotation,

$$R \in SO(3),$$

we have

$$R^T R = I, \quad \det(R) = 1.$$

Therefore,

$$R^{-T} = R.$$

The transformation rule becomes

$$(Ru) \times (Rv) = R(u \times v).$$

Thus rotations transform normals exactly like ordinary vectors.

This explains why rotation matrices are particularly convenient in 3D geometry and computer graphics.

Scaling and Surface Normals

Consider a non-uniform scaling

$$A = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}.$$

Then

$$A^{-T} = \begin{bmatrix} 1/s_x & 0 & 0 \\ 0 & 1/s_y & 0 \\ 0 & 0 & 1/s_z \end{bmatrix}$$

and

$$\det(A) = s_x s_y s_z.$$

Therefore,

$$(Au) \times (Av) = s_x s_y s_z A^{-T}(u \times v).$$

This demonstrates why simply transforming a surface normal by A can

Practical Computer Graphics Example

Suppose a surface has normal

$$n = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and is transformed by

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The inverse transpose is

$$A^{-T} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} 1/2 \end{bmatrix}$$

Conceptual Summary

The progression is

Dot Product $\longrightarrow$ Inner Product $\longrightarrow$ Function-Space Geometry

and separately,

Cross Product $\longrightarrow$ 3D Area + Normal + Orientation.

The inner product provides

length, angle, orthogonality, projection.

The cross product provides

area, normal direction, orientation.

Finally,

$$(Au) \times (Av) = \det(A)A^{-T}(u \times v)$$

explains the special treatment of normals in 3D transformations.

Key Takeaways

- 1 The dot product measures alignment between vectors.
- 2 Orthogonality is characterized by

$$\langle u, v \rangle = 0.$$

- 3 Inner products generalize Euclidean geometry to abstract vector spaces and function spaces.
- 4 Function-space inner products lead directly to Fourier analysis, orthogonal polynomials and approximation theory.
- 5 The cross product is a special operation in $\mathbb{R}^3$:

$$u \times v \perp u, v.$$

- 6 Its magnitude equals the parallelogram area:

$$\|u \times v\| = \|u\| \|v\| \sin \theta.$$

- 7 Under an invertible transformation,

$$(Au) \times (Av) = \det(A) A^{-T}(u \times v).$$

- 8 For rotations,

$$(Ru) \times (Rv) = R(u \times v)$$

18. Cramer's Rule

Consider the square linear system

$$Ax = b$$

where

$$A \in \mathbb{R}^{n \times n}, \quad \det(A) \neq 0.$$

Since $\det(A) \neq 0$, the system has a unique solution.

Cramer's Rule gives each component of the solution directly:

$$x_i = \frac{\det(A_i)}{\det(A)}, \quad i = 1, \dots, n$$

where A_i is obtained from A by replacing its i -th column with b :

$$A_i = [a_1, \dots, a_{i-1}, b, a_{i+1}, \dots, a_n].$$

Thus every unknown is expressed as a ratio of determinants.

Cramer's Rule for a 2×2 System

Consider

$$\begin{cases} ax + by = p, \\ cx + dy = q. \end{cases}$$

The matrix form is

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}.$$

The determinant is

$$\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Replacing the first column,

$$\Delta_x = \begin{vmatrix} p & b \\ q & d \end{vmatrix} = pd - bq.$$

Replacing the second column,

Example: Cramer's Rule

Solve

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

We have

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

First,

$$\det(A) = 2(-1) - 1(1) = -3.$$

For x ,

$$A_1 = \begin{bmatrix} 5 & 1 \\ 1 & -1 \end{bmatrix},$$

so

Why Cramer's Rule Works

Write the columns of A as

$$A = [a_1, \dots, a_n]$$

and suppose

$$Ax = b.$$

Then

$$x_1 a_1 + \dots + x_n a_n = b.$$

Replace a_i by b :

$$A_i = [a_1, \dots, a_{i-1}, b, a_{i+1}, \dots, a_n].$$

Using

$$b = \sum_{j=1}^n x_j a_j,$$

Advantages and Limitations of Cramer's Rule

Advantages

- Gives an explicit symbolic formula for each unknown.
- Useful for theoretical analysis.
- Useful for small systems.
- Shows the direct connection between determinants and solutions.

Limitation

For large numerical systems, Cramer's Rule is computationally inefficient.

Cramer's Rule is primarily theoretical, not a practical numerical solver.

For large systems, methods such as

Gaussian elimination, LU factorization, QR factorization
are generally preferred.

19. Change of Basis

A vector does not depend on its coordinates.

Coordinates depend on the chosen basis.

Let

$$\mathcal{B} = \{b_1, \dots, b_n\}$$

be the old basis and

$$\mathcal{C} = \{c_1, \dots, c_n\}$$

be a new basis.

Suppose

$$P = \left[\begin{array}{c|c|c|c} | & | & & | \\ c_1 & c_2 & \cdots & c_n \\ | & | & & | \end{array} \right]_{\mathcal{B}}.$$

That is, the columns of P are the new basis vectors expressed in old coordinates.

Then

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{C}}.$$

Therefore,

$$[x]_{\mathcal{C}} = P^{-1}[x]_{\mathcal{B}}.$$

The matrix P is called the **change-of-basis matrix**.

Example: Change of Coordinates

Consider the standard basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

and the basis

$$\mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The change-of-basis matrix is

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Suppose

$$[x]_{\mathcal{C}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Then

Change of Basis for Linear Transformations

Let

$$T : V \rightarrow V$$

be a linear transformation.

Suppose its matrix in basis $\mathcal{B}$ is

$$[T]_{\mathcal{B}} = A$$

and its matrix in basis $\mathcal{C}$ is

$$[T]_{\mathcal{C}} = A'.$$

If

$$P = [I]_{\mathcal{B} \leftarrow \mathcal{C}},$$

then

$$A' = P^{-1}AP.$$

What Quantities Are Preserved Under Similarity?

If

$$A' = P^{-1}AP,$$

then A and A' represent the same linear transformation in different bases. Similarity preserves important structural quantities:

$$\det(A') = \det(A)$$

$$\operatorname{tr}(A') = \operatorname{tr}(A)$$

and, most importantly,

$$\det(\lambda I - A') = \det(\lambda I - A).$$

Therefore, similar matrices have the same

- determinant,
- trace,
- characteristic polynomial,

Change of Basis and Diagonalization

A particularly useful basis is an **eigenbasis**.

If a matrix A has n linearly independent eigenvectors, place them into

$$P = \begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{bmatrix}.$$

Then

$$AP = P \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}.$$

Hence

$$\boxed{P^{-1}AP = D}$$

where

$$\boxed{D = \text{diag}(\lambda_1, \dots, \lambda_n)}.$$

20. Eigenvalues and Eigenvectors

Let

$$A \in \mathbb{R}^{n \times n}.$$

A nonzero vector

$$v \neq 0$$

is an **eigenvector** of A if there exists a scalar

$$\lambda \in \mathbb{R}$$

such that

$$Av = \lambda v.$$

The scalar λ is the corresponding **eigenvalue**. Thus an eigenvector is a special direction that is not rotated away from its span by the transformation. The transformation simply scales it:

$$v \mapsto \lambda v.$$

Geometric Interpretation of Eigenvectors

For a general vector x ,

$$Ax$$

may change both its magnitude and direction.

For an eigenvector v ,

$$Av = \lambda v.$$

Therefore,

$$\text{span}\{v\}$$

is invariant under A :

$$A(\text{span}\{v\}) \subseteq \text{span}\{v\}.$$

Geometrically:

- $\lambda > 1$: stretching
- $0 < \lambda < 1$: contraction
- $\lambda < 0$: reversal of direction plus scaling

Characteristic Equation

Starting with

$$Av = \lambda v,$$

we obtain

$$Av - \lambda v = 0.$$

Therefore,

$$(A - \lambda I)v = 0.$$

A nonzero solution v exists only if

$$A - \lambda I$$

is singular.

Hence

$$\det(A - \lambda I) = 0.$$

Example: Finding Eigenvalues

Consider

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The characteristic equation is

$$\det(A - \lambda I) = 0.$$

Thus

$$\det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = 0.$$

Therefore,

$$(2 - \lambda)^2 - 1 = 0.$$

Expanding,

$$\lambda^2 - 4\lambda + 3 = 0.$$

Finding the Eigenvectors

For

$$\lambda_1 = 3,$$

solve

$$(A - 3I)v = 0.$$

We obtain

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

Thus

$$x = y.$$

Frame Title

An eigenvector is

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For

$$\lambda_2 = 1,$$

we solve

$$(A - I)v = 0,$$

giving

$$x = -y.$$

Hence

$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Diagonalization of the Example

Using the eigenvectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

construct

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The corresponding eigenvalues are

$$\lambda_1 = 3, \quad \lambda_2 = 1.$$

Therefore,

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

We obtain

$$A = PDP^{-1}.$$

Equivalently,

$$P^{-1}AP = D.$$

The original transformation becomes a simple scaling transformation when expressed in the eigenbasis.

Eigenvalues and Powers of a Matrix

If

$$A = PDP^{-1},$$

then

$$A^k = (PDP^{-1})^k.$$

Using cancellation,

$$A^k = PD^kP^{-1}.$$

Since

$$D = \text{diag}(\lambda_1, \dots, \lambda_n),$$

we have

$$D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

This can make repeated matrix multiplication dramatically simpler.

Eigenvalues and Stability

Consider the discrete system

$$x_{k+1} = Ax_k.$$

If v is an eigenvector,

$$Av = \lambda v.$$

Starting with

$$x_0 = v,$$

we obtain

$$x_1 = \lambda v,$$

$$x_2 = \lambda^2 v,$$

and generally

$$x_k = \lambda^k v.$$

Therefore:

$$|\lambda| < 1 \Rightarrow \text{decay},$$

$$|\lambda| > 1 \Rightarrow \text{growth},$$

$$|\lambda| = 1 \Rightarrow \text{neutral behaviour}.$$

Thus eigenvalues determine the stability of many dynamical systems.

Applications of Eigenvalues

Eigenvalues and eigenvectors appear throughout applied mathematics:

- **Differential equations:**

$$x'(t) = Ax(t)$$

- **Vibration analysis:** natural frequencies and modes.
- **Stability analysis:** eigenvalues determine growth and decay.
- **Quantum mechanics:** observables and measurement states.
- **Data science:** principal component analysis.
- **Graph analysis:** spectral properties of networks.
- **Control theory:** system dynamics and stability.

The common idea is that eigenvectors reveal intrinsic directions associated with a transformation or operator.

Eigenvalues in Data Science: PCA

Let the covariance matrix of a dataset be

$$\Sigma.$$

Since Σ is symmetric,

$$\Sigma = \Sigma^T,$$

it has real eigenvalues and an orthogonal eigenbasis. Solve

$$\Sigma v_i = \lambda_i v_i.$$

The eigenvectors define principal directions. The corresponding eigenvalues measure the variance in those directions:

$$\lambda_i = \text{variance along principal direction } v_i.$$

Ordering

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$$

allows PCA to retain the directions containing the largest amount of variance.

Thus eigenvalues connect linear algebra directly with dimensionality

The Conceptual Connection

The three topics are deeply connected.

Cramer's Rule $\longrightarrow$ Determinants

Change of Basis $\longrightarrow$ Similarity $\longrightarrow$ Diagonalization

Eigenvectors $\longrightarrow$ Special Basis $\longrightarrow$ Simple Representation

In particular, diagonalization asks:

Can we find a basis in which the transformation is diagonal?

If yes,

$$A = PDP^{-1}.$$

The columns of P are eigenvectors and the diagonal entries of D are the corresponding eigenvalues.

Key Takeaways

- 1 Cramer's Rule solves

$$Ax = b$$
$$x_i = \frac{\det(A_i)}{\det(A)}.$$

- 2 Change of basis changes coordinates, not the underlying vector:

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{C}}.$$

- 3 A linear transformation changes matrix representation according to

$$[T]_{\mathcal{C}} = P^{-1}[T]_{\mathcal{B}}P.$$

- 4 Eigenvectors satisfy

$$Av = \lambda v.$$

- 5 Eigenvalues satisfy

$$\det(A - \lambda I) = 0.$$

- 6 If there are enough independent eigenvectors,

$$A = PDP^{-1}.$$

21. Eigenvectors

Let

$$A \in \mathbb{R}^{n \times n}.$$

A nonzero vector

$$v \neq 0$$

is called an **eigenvector** of A if

$$Av = \lambda v$$

for some scalar

$$\lambda \in \mathbb{R}.$$

The scalar λ is called the corresponding **eigenvalue**. Thus an eigenvector is a special direction that is preserved by the transformation.

$$v \mapsto Av = \lambda v.$$

The transformation may change the magnitude and possibly reverse the direction, but the vector remains in the same one-dimensional subspace:

$$\text{span}\{v\}.$$

Geometric Interpretation

For a general vector x ,

$$Ax$$

may point in a completely different direction from x .

For an eigenvector v ,

$$Av = \lambda v.$$

Therefore,

$$Av \in \text{span}\{v\}.$$

Hence

$\text{span}\{v\}$ is an invariant subspace of A .
--

The value of λ determines the action:

λ	Effect
$\lambda > 1$	stretching
$0 < \lambda < 1$	contraction
$\lambda = 1$	unchanged
$\lambda = 0$	collapse to zero
$\lambda < 0$	reversal + scaling

Thus eigenvectors reveal the intrinsic directions of a linear transformation.

Finding Eigenvectors

Starting from

$$Av = \lambda v,$$

we obtain

$$Av - \lambda v = 0.$$

Therefore,

$$(A - \lambda I)v = 0.$$

For a nonzero solution v to exist,

$$A - \lambda I$$

must be singular.

Hence

$$\det(A - \lambda I) = 0.$$

Eigenspace

For an eigenvalue λ , the associated **eigenspace** is

$$E_\lambda = \ker(A - \lambda I).$$

Equivalently,

$$E_\lambda = \{v \in \mathbb{R}^n : (A - \lambda I)v = 0\}.$$

Since

$$(A - \lambda I)v = 0$$

is a homogeneous linear system, E_λ is a subspace. It contains

$$\mathbf{0}$$

even though the zero vector is not itself an eigenvector. Thus

E_λ is the subspace consisting of all eigenvectors corresponding to λ together

Example: Eigenspaces

Consider

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = 3, \quad \lambda_2 = 1.$$

For $\lambda_1 = 3$,

$$A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Solving

$$(A - 3I)v = 0$$

gives

$$x = y.$$

Therefore,

$$E_3 = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

For $\lambda_2 = 1$,

$$A - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

so

$$x = -y.$$

Hence

$$E_1 = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

Eigenspaces are Subspaces

Because

$$E_\lambda = \ker(A - \lambda I),$$

the eigenspace automatically satisfies the subspace properties.
If

$$u, v \in E_\lambda,$$

then

$$Au = \lambda u, \quad Av = \lambda v.$$

Therefore,

$$A(u + v) = Au + Av = \lambda u + \lambda v = \lambda(u + v).$$

Hence

$$u + v \in E_\lambda.$$

Linear Independence of Eigenvectors

A fundamental theorem states:

Theorem

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Suppose

$$Av_i = \lambda_i v_i$$

for

$$\lambda_1, \dots, \lambda_k$$

distinct.

Then

$v_1, \dots, v_k$ are linearly independent.

Consequently, if an $n \times n$ matrix has n distinct eigenvalues, it automatically has n linearly independent eigenvectors.

This is sufficient for diagonalization.

Diagonalization Through Eigenvectors

Suppose

$$v_1, \dots, v_n$$

are linearly independent eigenvectors of A with

$$Av_i = \lambda_i v_i.$$

Construct

$$P = \begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{bmatrix}$$

and

$$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}.$$

Then

$$AP = PD.$$

Since P is invertible,

$$\boxed{P^{-1}AP = D.}$$

Equivalently,

$$\boxed{A = PDP^{-1}.}$$

Thus diagonalization is a change of basis to an eigenbasis.

Why Diagonalization is Useful

Suppose

$$A = PDP^{-1}.$$

Then

$$A^k = PD^kP^{-1}.$$

Since

$$D^k = \begin{bmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{bmatrix},$$

computing large powers of A becomes much easier.
Similarly, when defined,

$$e^{At} = Pe^{Dt}P^{-1}$$

where

$$e^{Dt} = \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t} \end{bmatrix}.$$

This is fundamental in solving systems of differential equations.

Applications of Eigenvectors

Eigenvectors and eigenvalues provide the mathematical structure behind:

- **Stability analysis**

$$x_{k+1} = Ax_k$$

- **Differential equations**

$$x'(t) = Ax(t)$$

- **Vibration analysis**

Eigenvectors represent natural modes of vibration.

- **Principal Component Analysis**

Eigenvectors of covariance matrices define principal directions.

- **Dynamical systems**

Eigenvalues determine growth, decay and stability.

- **Quantum mechanics**

Eigenvectors represent states associated with measurable quantities.

23. Descriptive Statistics I

Definition

Descriptive statistics is the branch of statistics concerned with organising, summarising, and describing the essential characteristics of observed data.

Let the observed sample be

$$x_1, x_2, \dots, x_n.$$

Descriptive statistics does not, by itself, make probabilistic claims about an unobserved population. Instead, it describes the information contained in the available sample.

Main tools

- Frequency distributions and histograms
- Measures of central tendency
- Measures of dispersion
- Order statistics

23. Descriptive Statistics II

- Covariance and correlation
- Covariance matrices for multivariate data

Central tendency

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The median is the middle order statistic after sorting the observations.

Dispersion

The sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The sample standard deviation is

23. Descriptive Statistics III

$$s = \sqrt{s^2}.$$

Descriptive statistics is fundamental in data science because it provides an initial understanding of scale, variability, skewness, outliers, and relationships among variables.

24. Histogram I

Definition

A histogram is a graphical representation of the empirical distribution of numerical data.

The observation space is divided into intervals called *bins* or *classes*.

Suppose the bins are

$$B_1, B_2, \dots, B_k.$$

Let

$$n_j = \#\{x_i : x_i \in B_j\}$$

be the number of observations falling in bin B_j .

The relative frequency is

$$f_j = \frac{n_j}{n}.$$

24. Histogram II

If all bins have equal width h , the histogram height may be proportional to the frequency

$$n_j$$

or relative frequency

$$\frac{n_j}{n}.$$

For unequal bin widths, the correct density height is

$$\text{height}_j = \frac{n_j}{n h_j},$$

where h_j is the width of bin j .

Therefore,

24. Histogram III

$$\sum_{j=1}^k (\text{height}_j) h_j = 1.$$

Interpretation

A histogram provides a visual approximation to the empirical distribution of the data.

It can reveal:

- centre,
- spread,
- skewness,
- modality,
- possible outliers,
- approximate distributional shape.

24. Histogram IV

Important limitation

The appearance of a histogram depends on the choice of bin width and bin boundaries.

A very small bin width may produce a noisy histogram, whereas a very large bin width may hide important structure.

A common rule for choosing bin width is Scott's rule:

$$h = \frac{3.5s}{n^{1/3}}.$$

Another widely used rule is the Freedman–Diaconis rule:

$$h = 2 \frac{\text{IQR}(x)}{n^{1/3}}.$$

25. Sample Mean and Variance I

Let

$$X_1, X_2, \dots, X_n$$

be observations from a population with mean μ and variance σ^2 .
The corresponding observed values are

$$x_1, x_2, \dots, x_n.$$

Sample mean

The sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

and, for observed data,

25. Sample Mean and Variance II

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The sample mean estimates the population mean μ .

Properties

If $X_1, \dots, X_n$ are independent and identically distributed (i.i.d.) with $E[X_i] = \mu$, then

$$E[\bar{X}] = \mu.$$

Hence $\bar{X}$ is an unbiased estimator of μ .

Furthermore,

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

25. Sample Mean and Variance III

Thus increasing the sample size reduces the variability of the sample mean.

Sample variance

The unbiased sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

For observed data,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The denominator is $n - 1$, rather than n , when estimating the population variance σ^2 from a sample.

Why $n - 1$?

Since

25. Sample Mean and Variance IV

$$\sum_{i=1}^n (X_i - \bar{X}) = 0,$$

only $n - 1$ deviations can vary independently once $\bar{X}$ has been determined. These are the *degrees of freedom*.

For i.i.d. observations with finite variance,

$$E[S^2] = \sigma^2.$$

Thus S^2 is an unbiased estimator of the population variance.

Computational identity

An equivalent formula is

$$S^2 = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n - 1}.$$

Example

25. Sample Mean and Variance V

Consider

$$x = \{2, 4, 4, 6, 8\}.$$

Then

$$\bar{x} = \frac{2 + 4 + 4 + 6 + 8}{5} = 4.8.$$

Therefore,

$$s^2 = \frac{(2 - 4.8)^2 + (4 - 4.8)^2 + (4 - 4.8)^2 + (6 - 4.8)^2 + (8 - 4.8)^2}{4}.$$

Hence

$$s^2 = 5.2$$

25. Sample Mean and Variance VI

and

$$s \approx 2.280.$$

26. Order Statistics I

Definition

Given a sample

$$X_1, X_2, \dots, X_n,$$

the order statistics are the observations arranged in nondecreasing order:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}.$$

Here $X_{(k)}$ denotes the k -th smallest observation.

The notation

$$X_{(k)}$$

does not refer to the original observation X_k ; it refers to its position after sorting.

Important order statistics

26. Order Statistics II

$$X_{(1)} = \min(X_1, \dots, X_n)$$

is the sample minimum, while

$$X_{(n)} = \max(X_1, \dots, X_n)$$

is the sample maximum.

The sample range is

$$R = X_{(n)} - X_{(1)}.$$

For a sample of size n , the median is

$$\text{Median} = \begin{cases} X_{(\frac{n+1}{2})}, & n \text{ odd,} \\ \frac{X_{(n/2)} + X_{(n/2+1)}}{2}, & n \text{ even.} \end{cases}$$

26. Order Statistics III

Quantiles

Order statistics form the basis of empirical quantiles.

The first and third quartiles are commonly denoted by

$$Q_1, \quad Q_3.$$

The interquartile range is

$$\boxed{\text{IQR} = Q_3 - Q_1.}$$

The IQR is robust to extreme observations and is therefore useful for outlier detection.

A common rule identifies observations outside

$$[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$$

as potential outliers.

26. Order Statistics IV

Distributional interpretation

If $X_1, \dots, X_n$ are i.i.d. with cumulative distribution function F , then the distribution of the k -th order statistic is

$$P(X_{(k)} \leq x) = \sum_{j=k}^n \binom{n}{j} [F(x)]^j [1 - F(x)]^{n-j}.$$

If F has density f , then the density of $X_{(k)}$ is

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x).$$

Applications

Order statistics are important in:

- robust statistics,
- percentile estimation,

26. Order Statistics V

- quantile regression,
- outlier detection,
- non-parametric statistics,
- ranking and recommendation systems,
- latency analysis in computer systems.

27. Sample Covariance I

Suppose two variables are observed on the same n samples:

$$X_1, \dots, X_n$$

and

$$Y_1, \dots, Y_n.$$

Their sample means are

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

Definition

The sample covariance between X and Y is

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}).$$

27. Sample Covariance II

For observed data,

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

Interpretation

The covariance measures the direction of linear co-variation.

$$S_{XY} > 0$$

suggests that large values of X tend to occur with large values of Y .

$$S_{XY} < 0$$

suggests that large values of X tend to occur with small values of Y .

$$S_{XY} \approx 0$$

27. Sample Covariance III

indicates little linear co-variation.

However,

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent}$$

in general.

Zero covariance rules out linear dependence, but nonlinear dependence may still exist.

Alternative computational form

$$S_{XY} = \frac{\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}}{n-1}.$$

Relation to correlation

Covariance depends on the units of measurement.

The correlation coefficient normalises covariance:

27. Sample Covariance IV

$$r_{XY} = \frac{S_{XY}}{S_X S_Y}.$$

Thus

$$-1 \leq r_{XY} \leq 1.$$

Correlation is dimensionless, whereas covariance is not.

Geometric interpretation

After centering the observations, define

$$\mathbf{x}_c = (x_1 - \bar{x}, \dots, x_n - \bar{x})^T$$

and

$$\mathbf{y}_c = (y_1 - \bar{y}, \dots, y_n - \bar{y})^T.$$

27. Sample Covariance V

Then

$$S_{XY} = \frac{1}{n-1} \mathbf{x}_c^T \mathbf{y}_c.$$

Therefore, sample covariance is a scaled dot product between centered data vectors.

This connects descriptive statistics directly with linear algebra.

28. Sample Covariance Matrix I

Consider p variables measured over n observations.
Represent observation i by

$$\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix} \in \mathbb{R}^p.$$

The sample mean vector is

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i.$$

Define the centered observation

$$\mathbf{z}_i = \mathbf{x}_i - \bar{\mathbf{x}}.$$

28. Sample Covariance Matrix II

Definition

The sample covariance matrix is

$$S = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T.$$

Equivalently,

$$S = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1p} \\ S_{21} & S_{22} & \cdots & S_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ S_{p1} & S_{p2} & \cdots & S_{pp} \end{bmatrix}.$$

The diagonal entries are sample variances:

$$S_{jj} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2.$$

28. Sample Covariance Matrix III

The off-diagonal entries are sample covariances:

$$S_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

Matrix formulation

Let the data matrix be

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix}.$$

Let X_c be the centered data matrix obtained by subtracting the column mean from every column.

Then

28. Sample Covariance Matrix IV

$$S = \frac{1}{n-1} X_c^T X_c.$$

This is one of the most important connections between statistics and linear algebra.

Properties

The covariance matrix is symmetric:

$$S = S^T.$$

It is also positive semidefinite. For every $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^T S \mathbf{a} \geq 0.$$

Indeed,

28. Sample Covariance Matrix V

$$\mathbf{a}^T S \mathbf{a} = \frac{1}{n-1} \sum_{i=1}^n \left[\mathbf{a}^T (\mathbf{x}_i - \bar{\mathbf{x}}) \right]^2 \geq 0.$$

Hence every eigenvalue of S is nonnegative.

Connection with PCA

Principal Component Analysis (PCA) solves

$$S \mathbf{v} = \lambda \mathbf{v}.$$

The eigenvectors of S determine principal directions, while the corresponding eigenvalues measure the variance captured in those directions.

Thus,

Covariance matrix $\longrightarrow$ Eigenvectors $\longrightarrow$ Principal components.

28. Sample Covariance Matrix VI

Applications

- PCA and dimensionality reduction
- Feature analysis in machine learning
- Computer vision
- Signal processing
- Portfolio and risk analysis
- Multivariate statistical modelling
- Sensor fusion and robotics

29. Frequentist Statistics and Sampling I

Frequentist viewpoint

In frequentist statistics, the population parameters are treated as fixed but unknown quantities, while the observed sample is regarded as a realization of a random sampling process.

For example,

$$X_1, \dots, X_n \sim \text{i.i.d. } P_\theta,$$

where θ is an unknown parameter.

The data are random before observation, whereas θ is fixed.

An estimator is a function of the sample:

$$\hat{\theta} = T(X_1, \dots, X_n).$$

After observing the data,

$$\hat{\theta}_{\text{obs}} = T(x_1, \dots, x_n).$$

29. Frequentist Statistics and Sampling II

Sampling

Sampling is the process of selecting observations from a population to obtain information about unknown population characteristics.

A common theoretical assumption is simple random sampling, often represented by

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} P_\theta.$$

This assumption means:

- 1 observations have the same distribution;
- 2 observations are mutually independent.

29. Frequentist Statistics and Sampling III

Sampling distribution

An estimator is itself a random variable before the data are observed.
For example,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

has sampling distribution

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

when the observations are normally distributed.

Even when the population is not normal, the Central Limit Theorem often gives, under suitable regularity conditions,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

29. Frequentist Statistics and Sampling IV

Population, sample and estimator

Population $\rightarrow$ Random Sample $\rightarrow$ Estimator $\rightarrow$ Inference.

Examples:

$$\mu \longrightarrow \bar{X},$$

$$\sigma^2 \longrightarrow S^2.$$

Frequentist inference uses the sampling distribution of estimators to quantify uncertainty.

30. Mean Square Error (MSE) I

Let θ be an unknown parameter and let

$$\hat{\theta}$$

be an estimator of θ .

Definition

The Mean Square Error (MSE) is

$$\text{MSE}(\hat{\theta}) = E \left[(\hat{\theta} - \theta)^2 \right].$$

MSE measures the expected squared discrepancy between an estimator and the true parameter.

Bias

The bias of $\hat{\theta}$ is

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

30. Mean Square Error (MSE) II

The fundamental bias–variance decomposition is

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

Derivation

Write

$$\hat{\theta} - \theta = \left(\hat{\theta} - E[\hat{\theta}] \right) + \left(E[\hat{\theta}] - \theta \right).$$

Squaring and taking expectations,

$$E[(\hat{\theta} - \theta)^2] = E[(\hat{\theta} - E[\hat{\theta}])^2] + (E[\hat{\theta}] - \theta)^2,$$

because the cross term has expectation zero.

Therefore,

$$\text{MSE} = \text{Variance} + \text{Bias}^2.$$

30. Mean Square Error (MSE) III

Unbiased estimator

If

$$E[\hat{\theta}] = \theta,$$

then

$$\text{Bias}(\hat{\theta}) = 0$$

and therefore

$$\boxed{\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}).}$$

Example: sample mean

For i.i.d. observations with mean μ and variance σ^2 ,

$$E[\bar{X}] = \mu$$

30. Mean Square Error (MSE) IV

and

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Hence

$$\boxed{\text{MSE}(\bar{X}) = \frac{\sigma^2}{n}.$$

As n increases,

$$\text{MSE}(\bar{X}) \rightarrow 0.$$

Machine learning interpretation

For prediction,

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

30. Mean Square Error (MSE) V

It is widely used as a regression loss function.

The statistical MSE of an estimator and the empirical prediction MSE are related concepts, but they should not be treated as identical quantities.

31. Consistency of Estimators I

An estimator should ideally become increasingly accurate as the sample size grows.

This motivates the concept of *consistency*.

Definition

An estimator sequence

$$\hat{\theta}_n$$

is consistent for θ if

$$\hat{\theta}_n \xrightarrow{P} \theta$$

as

$$n \rightarrow \infty.$$

The notation $\xrightarrow{P}$ denotes convergence in probability.

31. Consistency of Estimators II

Equivalently, for every $\varepsilon > 0$,

$$P\left(|\hat{\theta}_n - \theta| > \varepsilon\right) \longrightarrow 0.$$

Thus, with increasing sample size, the probability that the estimation error exceeds any fixed positive tolerance tends to zero.

Example: consistency of the sample mean

Suppose

$$X_1, X_2, \dots$$

are i.i.d. with

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

The Weak Law of Large Numbers states that

31. Consistency of Estimators III

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu.$$

Therefore,

$\bar{X}_n$ is a consistent estimator of μ .

A useful sufficient condition for consistency is

$$\text{MSE}(\hat{\theta}_n) \longrightarrow 0.$$

Indeed, by Markov's inequality,

$$P(|\hat{\theta}_n - \theta| > \varepsilon) \leq \frac{E[(\hat{\theta}_n - \theta)^2]}{\varepsilon^2}.$$

Hence

31. Consistency of Estimators IV

$$\text{MSE}(\hat{\theta}_n) \rightarrow 0$$

implies

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

Bias and consistency

An estimator can be biased and still be consistent.

For example, an estimator may satisfy

$$\text{Bias}(\hat{\theta}_n) \rightarrow 0$$

and

$$\text{Var}(\hat{\theta}_n) \rightarrow 0.$$

Then

31. Consistency of Estimators V

$$\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n) + \text{Bias}(\hat{\theta}_n)^2 \rightarrow 0,$$

which implies consistency.

Key distinction

Unbiasedness $\neq$ Consistency.

Unbiasedness concerns the expectation at a particular sample size, whereas consistency concerns asymptotic behaviour as

$$n \rightarrow \infty.$$

32. Confidence Intervals I

A confidence interval provides a procedure for constructing an interval that has a specified long-run coverage probability.

Let θ be an unknown population parameter.

A $(1 - \alpha)$ confidence interval is a random interval

$$[L(X), U(X)]$$

satisfying

$$P_{\theta}(L(X) \leq \theta \leq U(X)) = 1 - \alpha$$

under the assumed model, or approximately so when an approximation is used.

Important interpretation

A 95% confidence procedure means that, over repeated samples, approximately 95% of intervals constructed by that procedure would contain the fixed parameter θ .

32. Confidence Intervals II

It is not, in the standard frequentist interpretation, correct to say that after observing the data there is a 95% probability that the fixed parameter lies inside this particular interval.

Confidence interval for a population mean

If

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$$

and σ is known, then

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Therefore, a $(1 - \alpha)$ confidence interval is

$$\boxed{\bar{X} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}}$$

32. Confidence Intervals III

If σ is unknown, replace σ by S and use the Student's t distribution:

$$\bar{X} \pm t_{n-1, 1-\alpha/2} \frac{S}{\sqrt{n}}.$$

Here

$$t_{n-1, 1-\alpha/2}$$

is the $(1 - \alpha/2)$ quantile of the Student's t distribution with $n - 1$ degrees of freedom.

Effect of sample size

The standard error of the sample mean is

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

32. Confidence Intervals IV

Therefore, increasing n reduces the width of the confidence interval approximately at the rate

$$\frac{1}{\sqrt{n}}.$$

General structure

Many confidence intervals have the form

$$\text{estimate} \pm \text{critical value} \times \text{standard error}.$$

This provides a direct connection between point estimation and uncertainty quantification.

33. Parametric Model Estimation I

Definition

A parametric statistical model assumes that the distribution of the data belongs to a family indexed by a finite-dimensional parameter vector.

$$\mathcal{P} = \{P_\theta : \theta \in \Theta \subseteq \mathbb{R}^d\}.$$

Here,

$$\theta = (\theta_1, \dots, \theta_d)$$

is the unknown parameter vector.

Examples

Normal model:

$$X_i \sim N(\mu, \sigma^2).$$

Parameters:

33. Parametric Model Estimation II

$$\theta = (\mu, \sigma^2).$$

Bernoulli model:

$$X_i \sim \text{Bernoulli}(p),$$

with

$$\theta = p.$$

Poisson model:

$$X_i \sim \text{Poisson}(\lambda),$$

with

$$\theta = \lambda.$$

33. Parametric Model Estimation III

Likelihood

For independent observations, the likelihood is

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n f(x_i; \theta).$$

The maximum likelihood estimator (MLE) is

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} L(\theta; \mathbf{x}).$$

Equivalently, maximise the log-likelihood

$$\ell(\theta; \mathbf{x}) = \log L(\theta; \mathbf{x}) = \sum_{i=1}^n \log f(x_i; \theta).$$

Thus,

33. Parametric Model Estimation IV

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} \ell(\theta; \mathbf{x}).$$

Example: Bernoulli model

Suppose

$$X_i \sim \text{Bernoulli}(p).$$

Then

$$P(X_i = x_i) = p^{x_i} (1 - p)^{1-x_i}.$$

Hence,

$$L(p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}.$$

Therefore,

33. Parametric Model Estimation V

$$L(p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}.$$

The log-likelihood is

$$\ell(p) = \left(\sum x_i \right) \log p + \left(n - \sum x_i \right) \log(1 - p).$$

Differentiating,

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p}.$$

Setting this equal to zero gives

$$\hat{p}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}.$$

Method of Moments

33. Parametric Model Estimation VI

Another parametric estimation method equates sample moments to population moments.

For example,

$$E_{\theta}[X] = m_1(\theta)$$

is estimated by

$$\bar{X} = m_1(\theta).$$

For multiple parameters, multiple moment equations are used.

Advantages

- Compact model representation
- Often computationally efficient
- Enables likelihood-based inference
- Can provide highly efficient estimators when the model is correct

33. Parametric Model Estimation VII

Limitation

Parametric inference depends on the assumed model being appropriate. Model misspecification can lead to biased or misleading conclusions.

34. Non-Parametric Model Estimation I

Definition

Non-parametric statistical methods do not assume that the underlying distribution belongs to a fixed finite-dimensional parametric family.

The model complexity may grow with the amount of data.

The empirical cumulative distribution function (ECDF) is

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq x\}}.$$

Here

$$\mathbf{1}_{\{X_i \leq x\}} = \begin{cases} 1, & X_i \leq x, \\ 0, & X_i > x. \end{cases}$$

The ECDF provides a non-parametric estimate of the population distribution function $F(x)$.

34. Non-Parametric Model Estimation II

Kernel Density Estimation

A smooth non-parametric estimate of a probability density function is the kernel density estimator:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right).$$

Here:

- K is a kernel function;
- $h > 0$ is the bandwidth;
- n is the sample size.

34. Non-Parametric Model Estimation III

A common choice is the Gaussian kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}.$$

Bandwidth interpretation

The bandwidth h controls the smoothness of the estimate.

Small h :

$\Rightarrow$ low bias but high variance.

Large h :

$\Rightarrow$ higher bias but lower variance.

This is another manifestation of the bias–variance trade-off.

Examples of non-parametric methods

34. Non-Parametric Model Estimation IV

- Empirical distribution functions
- Kernel density estimation
- Histograms
- k-nearest-neighbour methods
- Rank-based tests
- Non-parametric regression
- Smoothing methods

Parametric vs non-parametric

Property	Parametric	Non-parametric
Model form	Specified family	Flexible
Parameters	Finite-dimensional	Potentially data-dependent
Assumptions	Stronger	Weaker
Flexibility	Lower	Higher
Interpretability	Often high	Often lower
Data requirement	Often lower	Often higher

34. Non-Parametric Model Estimation V

Neither approach is universally superior.

Parametric models can be highly efficient when correctly specified, whereas non-parametric methods provide greater flexibility when the distributional form is uncertain.

35. Unified Statistical Workflow I

The concepts in this module form a coherent statistical pipeline.

Data → Description → Estimation → Uncertainty → Inference

Step 1: Describe the data

Use

$$\bar{X}, \quad S^2, \quad X_{(k)}, \quad S_{XY}, \quad S.$$

These describe location, spread, ordering, and dependence.

Step 2: Understand sampling

Treat

$$X_1, \dots, X_n$$

as random observations generated by a population model.

Step 3: Construct estimators

35. Unified Statistical Workflow II

An unknown parameter

$$\theta$$

is estimated by

$$\hat{\theta} = T(X_1, \dots, X_n).$$

Step 4: Evaluate the estimator

Important criteria include

$$\text{Bias}(\hat{\theta}),$$

$$\text{Var}(\hat{\theta}),$$

$$\text{MSE}(\hat{\theta}),$$

35. Unified Statistical Workflow III

and consistency:

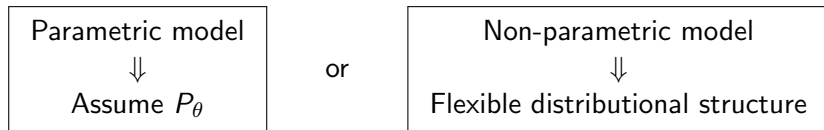
$$\hat{\theta}_n \xrightarrow{P} \theta.$$

Step 5: Quantify uncertainty

Construct confidence intervals such as

estimate $\pm$ critical value $\times$ standard error.

Step 6: Choose a modelling framework



Connection with machine learning

35. Unified Statistical Workflow IV

Many machine learning systems can be understood statistically as

Sample $\rightarrow$ Estimator $\rightarrow$ Loss/MSE $\rightarrow$ Generalisation $\rightarrow$ Uncertainty.

For multivariate data,

$$S = \frac{1}{n-1} X_c^T X_c$$

connects descriptive statistics with linear algebra and PCA.

Key conceptual chain

35. Unified Statistical Workflow V

Histogram → Distribution shape

Mean/Variance → Location and spread

Order Statistics → Quantiles and robustness

Covariance → Linear dependence

Covariance Matrix → Multivariate structure

Sampling → Sampling distributions

MSE → Estimator quality

Consistency → Asymptotic correctness

Confidence Intervals → Uncertainty quantification

Parametric/Non-parametric → Model estimation.

Module 3 — Summary I

Descriptive Statistics

- Histograms describe empirical distributional shape.
- The sample mean estimates population location.
- Sample variance measures dispersion.
- Order statistics provide minima, maxima, medians and quantiles.
- Covariance measures linear co-variation.
- The covariance matrix represents multivariate dependence.

Frequentist Statistics

- Sampling determines the distribution of estimators.
- MSE combines variance and squared bias.
- Consistency describes asymptotic convergence to the true parameter.
- Confidence intervals quantify uncertainty through repeated-sampling coverage.

Module 3 — Summary II

- Parametric estimation assumes a finite-dimensional model family.
- Non-parametric estimation provides greater distributional flexibility.

Central mathematical relationships

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

$$\hat{\theta}_n \xrightarrow{P} \theta$$

$$S = \frac{1}{n-1} X_c^T X_c$$

$$\text{Estimator} + \text{Sampling Distribution} + \text{Uncertainty} = \text{Statistical Inference}$$

Overall perspective

Descriptive statistics summarises observed data, while frequentist statistics uses probability and sampling distributions to make statements about unknown population parameters.

Together, these ideas provide a mathematical foundation for

Data Science	Machine Learning	Signal Processing	Computer Vision	En
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and establish the statistical framework required for more advanced topics such as hypothesis testing, regression, Bayesian inference, PCA, and statistical learning theory.